

EXECUTIVE SUMMARY REPORT

SELENIA Team J



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1.2 Applicable and Reference Documents

Applicable documents:	
AD01	DIMEAS/STAR/2026/001/SMSD/SOW/SMEX/I1R1 and its applicable documents
AD02	ESSB-ST-U-007, Issue 1 Rev. 1, Ott. 2025
AD03	ECSS-E-ST-10C Rev.1 – System engineering general requirements (15 February 2017)
AD04	ENGINEERING GUIDELINES FOR ESA IOD CUBESAT PROJECTS", European Space Agency, Manual/User Guide/Handbook ESA-TECSIU-MAN-2025-001198, Issue 1, revision 1, Mag. 2025.
AD05	United Nations, <i>United Nations Treaties and Principles on Outer Space</i> , New York: United Nations, 2002, Rep. ST/SPACE/11.
AD06	ECSS-M-ST-10C Rev.1 – Project planning and implementation (6March2009)
AD07	ECSS-E-ST-10-02C Rev.1 – Verification (1February2018)
AD08	ECSS-E-ST-10-03C Rev.1 – Testing (31May2022)
AD09	ECSS-E-HB-10-02A – Verification Guidelines (17December2010)
AD10	Arianespace, <i>Ariane 6 User's Manual</i> , Issue 2, Rev. 0, Arianespace, Evry-Courcouronnes, France, Feb. 2021.
AD11	Exolaunch, EXOpod Nova User Manual, Revision 1.2, June 2024.
AD12	Cal Poly, CubeSat Design Specification (CDS), Rev. 14.1, San Luis Obispo, CA.
Reference documents:	
RD01	S. V. Weston <i>et al.</i> , "State-of-the-Art Small Spacecraft Technology", NASA, Ames Research Center e Goddard Space Flight Center, Rep. NASA/TP—20250000142.
RD02	W. J. Larson and J. R. Wertz, Eds., <i>Space Mission Analysis and Design</i> , 3rd ed. El Segundo, CA: Microcosm Press.
RD03	NASA Space Systems Engineering Handbook, NASA/SP-2007-6105 Rev1
RD04	S. Corpino and F. Stesina, "Aerospace systems - AOCS," 2025.
RD05	S. Corpino and F. Stesina, "Aerospace systems - Communication system," 2025.

RD06	M. Samson, L. Kazemi, R. Van de Putte, and T. Delabie, "Attitude estimation strategies for a multi-star tracker ADCS," in <i>Proc. SPIE 13546, Small Satellites Systems and Services Symposium (4S 2024)</i> , 2025, Art. no. 135461H
RD07	ECSS-E-ST-60-30C - Satellite attitude and orbit control system (AOCS) requirements (30 August 2013)
RD08	E. Capello "Spacecraft Dynamics and Control – Actuators for position and attitude control", 2025.
RD09	Wertz, J. R., Everett, D. F., and Puschell, J. J., <i>Space Mission Engineering: The New SMAD</i> , Microcosm Press, 2011.
RD10	Curtis, H. D., <i>Orbital Mechanics for Engineering Students</i> , 4th ed., Elsevier, 2020.
RD11	Plíce, L., Galal, K., and Burns, J. O., "DARE Mission Design: Low RFI Observations from a Low-Altitude Frozen Lunar Orbit," 2017.
RD12	AGI / Ansys, <i>Systems Tool Kit User's Guide / STK Documentation</i> .
RD13	NASA, <i>Risk Management Handbook</i> , NASA/SP-2011-3422, 2011.
RD14	Lutz,D.(2023). The non-human persona guide.
RD15	J. W. Dankanich, J. Cardin, A. Dien, H. Kamhawi, C. J. Netwall, and M. Osborn, "Advanced Xenon Feed System (AXFS) Development and Hot-fire Testing," in <i>45th AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit</i> , Denver, CO, USA, Aug. 2009, AIAA 2009-4910.
RD16	K. A. Polzin and S. Peeples, "Iodine Hall Thruster Propellant Feed System for a CubeSat," in <i>50th AIAA/ASME/SAE/ASEE Joint Propulsion Conference</i> , Cleveland, OH, USA, Jul. 2014, AIAA 2014-3915.
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RD18	P. C. Lai, D. C. Sternberg, R. J. Haw, E. D. Gustafson, P. C. Adell, and J. D. Baker, "Lunar Flashlight CubeSat GNC System Development," <i>Acta Astronautica</i> , vol. 173, pp. 425–441, 2020.
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RD21	V. Muralidharan, M. R. Makhdoomi, A. Žinys, B. Razgus, M. Klimavičius, M. A. Olivares-Mendez, and C. Martinez, "On-Ground Validation of Orbital GNC: Visual Navigation Assessment in Robotic Testbed Facility," <i>Astrodynamics</i> , 2024.
RD22	R. Saylor, "Lunar Reconnaissance Orbiter (LRO) Mission Concept of Operations," NASA Goddard Space Flight Center, Greenbelt, MD, USA, Tech. Rep. 431-PLAN-000052, 2005.
RD23	T. A. Collard, J. P. Sheehan, and A. D. Gallimore, "Pressurized Xenon Propellant Management System for the CubeSat Ambipolar Thruster," in <i>Joint Conference of 30th ISTS, 34th IEPC and 6th NSAT</i> , Hyogo-Kobe, Japan, Jul. 2015, IEPC-2015-364.
RD24	Sekine, H., Toma, K., Matsumoto, T., Nakasuka, S., Tanaka, T., Ebinuma, T., Sobukawa, R., & Aoyanagi, Y. (n.d.). <i>Engineering Model Development of LunaCube: 6U Dual-satellite Lunar Navigation and Communication System</i>
RD25	S. Corpino, "Space Mission and systems design – Space Mission Development",2026
RD26	S. Corpino, "Space Mission and systems design – Mission Design",2026
RD27	ESA-FlatSat-for-CubeSat testing [2021]
RD28	CubeSat Engigneering-AIT Model Philosophy
RD29	ECSS-E-ST-70-11C Rev.1 – Space Segment Operability
RD30	Cappelletti, Chantal, et al. <i>CubeSat Handbook: From Mission Design to Operations</i> . London ; San Diego, Ca, Academic Press, 2021.
RD31	Zaman, Fahad, et al. "Absorbed Doses from GCR and Albedo Particles Emitted by the Lunar Surface." <i>Acta Astronautica</i> , vol. 175, 1 Oct. 2020, pp. 185–189.

RD32	Adams, J H, et al. "The Ionizing Radiation Environment on the Moon." <i>Advances in Space Research</i> , vol. 40, no. 3, 1 Jan. 2007, pp. 338–341.
RD33	DSMZ, <i>Data Sheet: Bacillus subtilis DSM 10</i> , Leibniz Institute DSMZ-German Collection of Microorganisms and Cell Cultures.
RD34	DSMZ, <i>Data Sheet: Methanosarcina barkeri DSM 800</i> , Leibniz Institute DSMZ-German Collection of Microorganisms and Cell Cultures.
RD35	DSMZ, <i>Data Sheet: Schizosaccharomyces pombe DSM 70572</i> , Leibniz Institute DSMZ-German Collection of Microorganisms and Cell Cultures.
RD36	DSMZ, <i>Data Sheet: Anabaena variabilis (Trichormus variabilis) ATCC 29413</i> , Leibniz Institute DSMZ-German Collection of Microorganisms and Cell Cultures.
RD37	https://wrmiss.org/workshops/twentyfourth/Berger_RAMIS.pdf

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5 Definitions and abbreviations

5.1 Definitions

N/A Not Applicable

5.2 Abbreviations

AD	Applicable Document
ADCS	Attitude and Determination Control System
AHP	Analytic Hierarchy Process
AOCS	Attitude and Orbit Control System
APE	Attitude Pointing Error
APR	Array Power Regulator
BLSS	Bioregenerative Life Support Systems
BTC	Blue Canyon Technologies
CAD	Computer-Aided Design
CAN	Controller Area Network
ConOps	Concept of Operations
COSPAR	Committee on Space Research
COTS	Commercial Off-The-Shelf
CoM	Center of Mass
CSS	Coarse Sun Sensor
CVCM	Collected Volatile Condensable Materials
DB	Data Budget
DC	Direct Current
DM	Digital Model
DoD	Depth of Discharge
EAI	European Astrobiology Institute
ECAPS	Ecological Advanced Propulsion Systems
ECSS	European Cooperation for Space Standardization
EIRP	Effective Isotropic Radiated Power
EKF	Extended Kalman Filter
EM	Engineering Model
EMI	Electromagnetic Interference
EOL	End Of Life
EOS	End Of Study
EPS	Electrical Power System
ESA	European Space Agency
FOMs	Figure of Merit
FOV	Field of View
FM	Flight Model
FR	Final Review
FSS	Fine Sun Sensors
FUN	Functional
GCR	Galactic Cosmic Rays
GNC	Guidance Navigation and Control
HK	Housekeeping
HLCL	Heater Latching Current Limiter

HLR	High Level Requirements
ICRP	International Commission on Radiological Protection
IMU	Inertial Measurement Unit
KOM	Kick-Off Meeting
LCL	Latch-up Counter Limiter
LET	Linear Energy Transfer
LEOP	Launch and Early Orbit Phase
LLO	Low Lunar Orbit
LMP	Liquid Monopropellant
LTC	Landmark Tracking Camera
MBSE	Model-Based Systems Engineering
MCU	Microcontroller Unit
MD	Mission Driver
MDR	Mission Definition Review
MIS	Mission
MK-7	Menaquinone-7
MLI	Multilayer Insulation
MO	Mission Objectives
MS	Mission Statement
MR	Mission Requirements
OBC	On Board Computer
OBDH	On Board data handling
OBS	Organisation(al) Breakdown Structure
OD	Optical Density
OM	Operative Modes
OPE	Operative
PCDU	Power Conditioning and Distribution Unit
PCU	Power Control unit
PDR	Preliminary Design Review
PDU	Power Distribution Unit
PFM	ProtoFlight Model
PHYS	Physical
PRO	Propulsion
QM	Qualification Model
R	Requirement
RCS	Reaction Control System
RD	Reference document
SEL	Single Event Latchup
SH	Stakeholder
SK	Station Keeping
SMSD	Space Missions and Systems Design
SO	Science Objectives
SoC	State of Charge
SoW	Statement of Work
SPE	Solar Particle Event
STK	System Tool Kit
STM	Science Traceability Matrix
TCS	Thermal Control System

TML	Total Mass Loss
TT&C	Telemetry, Tracking and Command
UART	Universal Asynchronous Receiver-Transmitter
UHF	Ultra-High Frequency
UNOOSA	United Nations Office for Outer Space Affairs
VHF	Very High Frequency
WBS	Work Breakdown Structure
WP	Work Package
WPD	Work Package Description

6 Abstract

This document presents the comprehensive mission and system concept developed during Phase 0/A of the SELENIA project, conducted by Team J at Politecnico di Torino within the Space Missions and Systems Design framework.

The primary engineering problem addressed is the complex and multidisciplinary optimization required to operate a highly constrained deep-space SmallSat in the harsh high-radiation cislunar environment, ensuring high scientific return while complying with rigid mass, volume, and programmatic constraints.

The overall methodology applied throughout this Phase 0/A study adheres strictly to ECSS standards, systematically transitioning from user-defined stakeholder requirements to a fully traceable functional architecture.

The engineering process initiated with a meticulous characterization of the lunar orbital environment, specifically evaluating a cumulative TID and predicting critical SEE over the lifetime. This environmental assessment directly guided the trade-off analyses for the primary spacecraft bus configurations, trajectory designs, and payload selection. Trajectory verification and station-keeping strategies were thoroughly integrated to validate long-term orbital stability, accounting for both nominal operational windows and robust off-nominal safety recovery profiles to ensure mission survivability.

The rigorous systems engineering and subsystem-level design activities successfully consolidated a robust, flight-feasible architecture, yielding the following core technical and programmatic results:

- **Space Segment and Platform Sizing:** A modular 12U CubeSat platform configuration has been selected and frozen, establishing a maximum wet mass baseline of 24.0 kg (including a 20% system margin) and standard stowed dimensions of 226.3 x 226.3 x 366.0 mm. The structural baseline is engineered to withstand severe launch-induced mechanical loads while accommodating a highly integrated, radiation-tolerant primary payload dedicated to astrobiological and environmental data collection.
- **Subsystem Sizing and Budget Consolidation:** Full technical closure has been demonstrated across all primary system budgets. The Electrical Power Subsystem (EPS) utilizes a high-efficiency, deployable solar array network capable of generating up to 85 W of electrical power, coupled with an advanced 120 Wh energy storage unit to sustain peak payload operations during eclipses. The Telemetry, Tracking, and Command (TT&C) configuration integrates deep-space UHF-band transponders with low-gain antenna arrays to avoid the need for high accuracy pointing, ensuring a positive link margin (>3 dB) at the nominal downlink data rate at lunar distance.
- **Mission Design and Trajectory Profile:** A comprehensive nominal timeline encompassing launch, cislunar transit, orbital insertion, and a dedicated operational phase designed to maximize spatial and temporal coverage of the target lunar regions. The nominal mission lifetime is constrained to less than 60 days to mitigate radiation hazards, while the EOL strategy incorporates an independent disposal trajectory sequence that ensures safe orbital removal, in compliance with COSPAR planetary protection guidelines.
- **Programmatic, Cost, and Risk Assessment:** The total lifecycle cost estimation for the Phase 0/A baseline is projected within the allocated industrial targets for deep-space SmallSat demonstrators. A comprehensive Risk Register identified critical areas, mainly radiation-induced component degradation and deep-space communications, which have been systematically mitigated down to an acceptable green/yellow criticality matrix through hardware redundancy, software-implemented fault tolerance, and strategic component selection.

Given the high fidelity of the technical closure, positive budget margins, and programmatic compliance achieved in this report, the SELENIA project is ready to proceed into the subsequent detailed design phases (Phase B) to further mature the payload interfaces, refine the attitude guidance algorithms, and proceed with hardware-in-the-loop breadboarding.

7 Scope and Purpose

This document summarizes the activities carried out by Team J – SELENIA within the project work of the 2026 SMSD class at Politecnico di Torino, in relation to the study of a small mission for cislunar radiation mapping and data collection.

The document is organized as follows. Section 1 reports the Document Control Data, including the Document Change Log, the Record of all changes and updates made throughout the design process, and the Applicable and Reference Documents. Sections 2 through 5 report the Table of Contents, List of Figures, List of Tables, and Definitions and Abbreviations, respectively. Section 6 presents the Abstract, summarizing the mission concept, study methodology, and main results. Section 7 defines the scope and purpose of the document. Section 8, Context and Objectives of the Mission, describes the beginning of the design process, from problem formulation to identifying mission drivers and developing the Science Traceability Matrix. Section 9 presents the mission design, covering mission requirements, Concept of Operations, mission architecture including the C3 communication architecture, mission analysis results comprising trajectory design and ΔV budget, and the mission-level risk assessment. Section 10 presents the final results of the overall system design, including the outcomes of individual subsystem analyses, CAD model, and mass and volume budget, as well as payload design for LunaRAY and LunaCELL, including microorganism selection and data budget, and the detailed design of all spacecraft subsystems: Command and Data Handling, Communication System, Electrical Power System, Attitude Determination and Control System, Guidance Navigation and Control, Thermal Control System, and Structural and Mechanism Subsystem. Section 11 covers the system development plan, comprising the Assembly, Integration, and Verification Plan – including the verification strategy, model philosophy, and AIV schedule – as well as the Capella implementation, which describes the MBSE workflow and its application. Section 12 covers study management, including the Work Breakdown Structure, study logic and timeline, team organization, reporting and communication infrastructure, project-level risk assessment, and cost estimation. Section 13 presents the conclusions and recommendations for the next design phase.

This document represents the final deliverable D2 for the study and will be used to assess the activities carried out by the team. It may also serve as a reference for any future study iteration.

8 Context and objectives of the mission

8.1 Problem Formulation

The establishment of sustainable human presence on the Moon is currently limited by the incomplete characterization of the lunar radiation environment and by the limited knowledge of its effects on biological systems. In particular, the contribution of Galactic Cosmic Rays, Solar Particle Events, and secondary radiation generated by the lunar surface must be quantified as a function of solar activity. Furthermore, the response of microorganisms employed in Bioregenerative Life Support Systems to extended exposure to space radiation remains largely unexplored. The lack of data limits the possibility of accurately assessing astronaut health risks and the development of reliable life-support technologies for long-duration lunar missions. Consequently, dedicated studies are required to simultaneously investigate the radiation field in the vicinity of the Moon and the biological effects induced by such an adverse environment.

8.2 Stakeholders Needs

From the SH analysis we highlighted primary needs and objectives of each stakeholder, and the results are summarised in Table 8-1. It is worth noting that not all SHs were included in the needs analysis: some hold too little power to be relevant to the production of mission values through the QFD matrix, but we decided to include them in Figure 8-1 to highlight the result of the preliminary SH analysis.

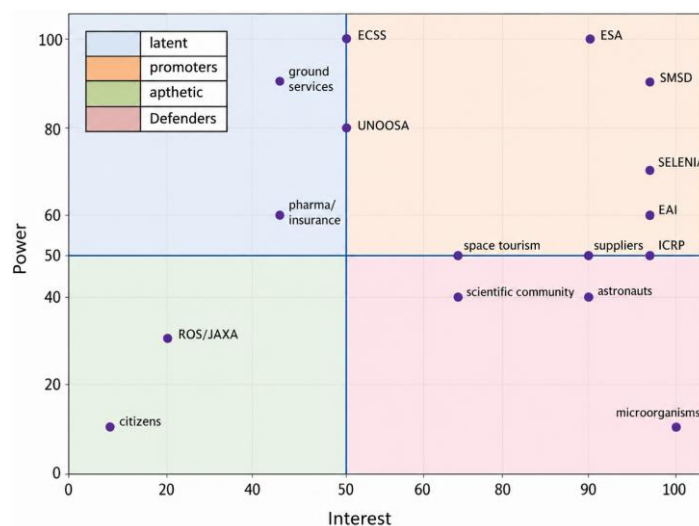


Figure 8-1: Stakeholders Power-Interest matrix

Table 8-1: Needs

Stakeholder	Needs
SMSG Advisory Board	System reliability; low outage probability; compliance with the project schedule; prediction of radiation exposure effects; compliance with space regulations; promotion of European cooperation
ESA	High-accuracy radiation measurements; high temporal and spatial resolution data; long-term radiation monitoring; prediction of radiation exposure effects; support for future lunar habitats; low development and operational costs; low outage probability.
Astronauts (EAC – JSP)	Reliable radiation exposure predictions and data supporting future human lunar exploration; high system reliability and operational safety.
ECSS	Compliance with engineering standards; data standardization; implementation of safety and redundancy principles.
UNOOSA	Compliance with international space regulations; environmentally responsible mission operations and disposal.
ICRP	High accuracy radiation measurements; long-term radiation monitoring; prediction of radiation exposure effects; data supporting astronaut protection strategies and future lunar habitats.
EAI	Biological response data under lunar radiation conditions; support for radiobiology studies and future bioregenerative life-support systems.
Ground service providers	Reliable communications and data transmission; compatibility with ground infrastructure and operational standards; high data availability and integrity; low outage probability.

Suppliers (D-Orbit, Thales Arcsec, Argotec)	Validation and advancement of radiation-resistant components and technologies; commercial opportunities.
Citizens	Environmentally responsible mission operations and mitigation of launch-related impacts.
Space tourism companies	Radiation environment characterization for commercial human missions; reduction of radiation risks for operators and passengers; support for future commercial lunar activities.

8.3 Mission Statement and Objectives

The mission statement is highlighted here:

“Our mission aims to characterize the lunar radiation environment and monitor the metabolic response of biological organisms in extreme conditions, to support the development of reliable systems, and the promotion of long-term human space exploration.”

The mission objectives are categorized as follows:

- Scientific Objectives:
 - To quantify and model the components of the lunar radiation environment and their variability
 - To evaluate the biological response of BLSS-relevant organisms at different radiation regimes to support the optimization of future lunar habitat’s design
 - To assess long-term implications of radiation exposure on regenerative Life-Support Systems and generative predictive models for human exploration scenarios
- Engineering Objectives:
 - To ensure system's reliability, radiation robustness and experiment replicability
 - To collect high quality and non-corrupted datasets, and ensure secure transmission

8.4 Mission Drivers

Table 8-2 presents the mission drivers identified for the project. These drivers have been derived from the mission objectives and serve as high-level design guidelines for the development process throughout the mission design phase. Among them, **D-MIS-04-00** and **D-MIS-05-00** are particularly representative of the project's vision, highlighting the emphasis placed on the reliability and scientific value of the generated data, as well as the innovative capabilities and advancements that the mission seeks to achieve.

Table 8-2: Mission Drivers

Driver Identifier	Driver Text
D-MIS-01-00	The mission shall preferably use European technologies.
D-MIS-02-00	The Mission should reuse as much as possible existing Earth infrastructures, ground antennas and logistics for operations.
D-MIS-03-00	Existing off-the-shelf products and technologies for both Ground and Space Segments should be selected when possible.
D-MIS-04-00	The mission should provide useful scientific data to support the development of technologies and mitigation strategies for future long-duration human habitat missions on the Moon.
D-MIS-05-00	The system should be designed to guarantee consistent transmission and preservation of scientific data, for the whole duration of the mission.

8.5 Science Traceability Matrix

Following the identification of the mission stakeholders, a Science Traceability Matrix (STM) was developed to translate stakeholder needs into scientific goals, measurable objectives, and technical requirements. The matrix links MDs to the resulting instrument and system specifications, ensuring traceability of the design process.

Only SHs with the greatest interest in the mission's scientific outcomes were considered. Their needs were associated with the outputs generated by the two payloads, LunaRAY and LunaCELL. In this way, SH expectations were progressively translated into scientific goals and objectives, defining the physical quantities to be measured and the corresponding performance requirements. These requirements were then used to derive instrument characteristics and mission-level constraints, whose impact on spacecraft architecture and subsystem sizing is captured in the Derived Requirements column. Finally, the expected outputs of each instrument are identified as Data Products.

For the radiation-related objectives, the needs of stakeholders such as ESA and ICRP were linked to measurements from LunaRAY. This led to the definition of requirements on quantities such as count rate, energy deposition, measurement range, and resolution, together with the associated mission-level constraints. Similarly, the interests of EAI were translated into biological objectives addressed by LunaCELL, leading to requirements on experiment repeatability and payload operations. The resulting STM establishes a direct connection between stakeholder needs, scientific objectives, technical requirements, and the data products generated by the mission.

Table 8-3: Science Traceability Matrix (only a section of the matrix, developed from ESA and EAI's needs, is shown)

SCIENCE TRACEABILITY MATRIX									
Stakeholder	Science need/question	Science goal	Science objective	Science requirements	Instrument/Technique	Measurements requirements	Instruments requirements	Derived requirements	Data products
European Space Agency (ESA)	Q1. How is the energy spectrum characterized in the radiative environment of lunar orbit? Q2. How does the spectrum vary with orbital radius (distance from the lunar surface) and with solar activity? How does the effect of radiation differ between less shielded and more shielded positions along orbit? Q3. To what extent can the hazard to electronic systems in lunar orbit be assessed?	Evaluate the quantity of radiative particles interacting with a system in lunar environment, and study their evolution along the operational orbit and in the long term.	Measure the count rate and the Energy Deposition of Solar Particle Events (SPEs), Galactic Cosmic Rays (GCRs) and Albedo Particles under different conditions, during the operational phase of the mission.	Particles shall be detected with an Energy Deposition Range: 0.06 MeV - 200 MeV.	Silicon Diodes (LunaRAY Detector)	General count rate time resolution shall be ≤ 1 minutes.	Energy resolution shall be better or equal to 40 energy bins per decade.	- The operational orbit shall be a quasi-frozen orbit with nominal altitudes of 100 km $\times$ 1500 km. - The spacecrafts shall be maintained in the selected quasi-frozen lunar orbit for the full nominal science phase, using station-keeping maneuvers. - The spacecraft shall maintain the operational orbit for at least 2 years, using a low-ΔV station-keeping strategy. - Detector shall be operative since after the orbit immersion is completed, during the whole mission, and shall end before disposal phase starts. - The spacecraft shall maintain a nadir-pointing attitude with respect to the Moon during all nominal science phases in the operational orbit.	Count-Rate and Energy deposition spectra, with timestamps
		Distinguish the interactions between Galactic Cosmic Rays (GCRs), Solar Particle Events (SPEs), and albedo particles, and evaluate the weight of each of them. Measure albedo particles quantity variation as a function of the distance from the Moon. Study the evolution of the detected energy per unit of time during orbital operations and over the long term.	Measure the Linear Energy Transfer (LET) of SPEs, GCRs and albedo particles throughout the operational phase of the mission.	The Linear Energy Transfer spectra to be detected shall be in the region: 0.1 - 250 keV/ μ m in water.		Energy Deposition Time Resolution shall be ≤ 5 minutes.	After the beginning of the experiment, The Diode shall be maintained in a temperature range of -20°C and 40°C.		LET spectra, with timestamps
European Astrobiology Institute (EAI)	Q1. How does radiation affect the proliferation of cellular microorganisms in lunar orbit? For how long can microorganisms survive in this environment? Q2. Is there a relationship between absorbed dose and the rate of microbial proliferation? Q3. How does variability in external conditions affect biological stability? How does it vary with solar activity? How does the proliferation of microorganisms differ between shielded and unshielded environments?	Analyze the evolution of populations of different types of microorganisms over time throughout the entire operational phase of the mission. Assess the growth rate of microorganisms population in the cuvettes. Compare the proliferation trends of microorganisms in shielded and unshielded environments.	Detect the Optical Density for every culture in regular time intervals	- Four different species shall be used. - Science operation shall be carried out on an unshielded culture and a shielded culture. - Three different media shall be used. - Every medium shall be pumped in one empty cuvette, as a control unit.	Photodiode (Optical Acquisition System, LunaCELL)	- Measurements shall start after cuvettes get filled by media. - Measurements shall happen with a rhythm of at least 2 x per 24 h.	- The diodes shall measure the OD in the cuvettes at 490 nm or 600 $\pm$ 10 nm. - The diodes shall measure light absorbance between 0.025 to 3 with an accuracy of 0.03 or better. - Thermal control subsystem shall maintain LunaCELL within a temperature range of +10°C and +40°C while storing the microorganisms in dried form.	- Scientific operations shall begin no later than 14 days after the immersion in the operational lunar orbit. - Photodiodes shall be operative since the activation of the Payload, during the whole mission, until the end of its operational lifetime, except during Survival Mode. - The measurements shall be validated through repeatability of the experiments in similar conditions in at least three distinct laboratories.	Optical Density measurements with timestamps

9 Mission Design

The mission design establishes the baseline lunar mission concept and provides the system-level rationale for the selected operational, architectural, and trajectory solutions. The design is driven by the need to ensure access to the lunar environment, maintain a stable operational orbit for scientific exposure, manage a four-spacecraft configuration, provide recurrent communication opportunities, limit the station-keeping demand, and guarantee controlled end-of-life disposal.

The selected baseline consists of four spacecraft deployed in lunar orbit by a mothercraft and operated on a common quasi-frozen lunar orbit. The mission profile includes orbit acquisition, nominal science operations, station keeping, and disposal. The adoption of a common reference orbit allows the trajectory design, manoeuvre logic, and operations concept to be standardized across the space segment, while the multi-spacecraft architecture improves measurement continuity and overall mission resilience.

9.1 Mission Requirements

The mission requirements were derived from the SH needs, the MO, and the main design drivers identified during the preliminary design phase. They define the system-level constraints and performance targets driving the selection of the operational concept, mission architecture, trajectory design, and disposal strategy. The requirements retained in this section are limited to the mission-design drivers directly affecting orbit selection, manoeuvre planning, operations, communication availability, and end-of-life compliance.

In accordance with the applicable ECSS standards [AD3] and ESA policy [AD11], each requirement was assigned a unique identifier and linked to its corresponding source. Table 9-1 reports the mission-design High-Level Requirements and the corresponding Mission Requirements retained for the baseline definition.

Table 9-1: Key Mission-Design Requirements

Requirement ID	Requirement	Type	Verification method
R-MIS-01-00	The mission shall be operative with only one launch	MIS	Review
R-MIS-02-00	The mission shall be operative by 2030.	MIS	Review
R-MIS-03-00	The spacecraft shall be designed for insertion into lunar orbit by a lunar orbiter	MIS	Review, Analysis
R-MIS-05-00	The mission's operational orbit shall be a quasi-frozen orbit with nominal altitudes of 100 km × 1500 km.	MIS	Review, Analysis
R-MIS-06-00	The spacecraft shall perform a transfer from the initial lunar orbit to the operational quasi-frozen orbit.	MIS	Review, Analysis
R-MIS-07-00	The initial orbital geometry shall allow the periselenium altitude to decrease towards the target value without requiring a dedicated periselenium-lowering manoeuvre.	MIS	Review, Analysis
R-MIS-08-00	The spacecraft shall be maintained in the selected quasi-frozen lunar orbit for the full nominal science phase using station-keeping manoeuvres.	MIS	Review, Analysis
R-MIS-10-00	The orbit maintenance strategy shall limit the cumulative station-keeping Delta-V within the available propulsion budget.	MIS	Review, Analysis
R-MIS-11-00	The selected orbit shall provide recurrent access opportunities with the mothercraft and/or ground segment for command uplink and science data downlink.	MIS	Review, Analysis
R-MIS-16-00	The mission timeline shall be designed so that the major Earth eclipse event does not occur during the nominal science phase.	MIS	Review, Analysis
R-MIS-19-00	At the end of the operational phase, the spacecraft shall execute a disposal sequence to leave the operational orbit.	MIS	Review, Analysis
R-MIS-20-00	The disposal sequence shall include an aposelenium-raising manoeuvre followed by a manoeuvre targeting a final eccentricity close to zero.	MIS	Review, Analysis
R-MIS-21-00	The launcher of the spacecraft shall be ESA Ariane 6.	MIS	Review
R-MIS-22-00	The spacecraft shall maintain a nadir-pointing attitude with respect to the Moon during all nominal science phases in the operational orbit.	MIS	Review, Analysis

The retained requirements introduce the main constraints driving the mission design baseline. R-MIS-01-00 and R-MIS-02-00 define the high-level programmatic boundaries, imposing compatibility with a single launch opportunity and mission operability by 2030. R-MIS-03-00 establishes the mission injection concept, requiring the spacecraft to be

delivered to the lunar environment by a lunar orbiter, thus directly affecting the deployment strategy and the initial orbital conditions.

The trajectory design is mainly driven by R-MIS-05-00, R-MIS-06-00 and R-MIS-07-00. These requirements define the transfer from the initial lunar orbit to the selected 100 km × 1500 km quasi-frozen operational orbit and impose the exploitation of the natural periselenium evolution to reduce the need for a dedicated periselenium-lowering manoeuvre. Orbit maintenance is instead governed by R-MIS-08-00, R-MIS-08-02 and R-MIS-10-00, which require station-keeping throughout the nominal science phase, define the ±5 km periselenium control band, and constrain the cumulative maintenance Delta-V within the available propulsion budget.

Operational feasibility is driven by R-MIS-11-00 and R-MIS-16-00. The selected orbit shall provide recurrent access opportunities with the mothercraft and/or ground segment for command uplink and science data downlink, while the mission timeline shall prevent the major Earth eclipse from occurring during the nominal science phase. Finally, R-MIS-19-00 and R-MIS-20-00 define the end-of-life strategy, requiring each spacecraft to leave the operational orbit through a disposal sequence composed of an aposelenium-raising phase followed by an eccentricity-reduction manoeuvre. These requirements ensure that disposal is treated as an integral mission-design constraint rather than as a post-operational activity.

9.2 Concept(s) of Operations

The Concept of Operations was developed by dividing the mission into a sequence of operational phases, from launch to end-of-life disposal. Each phase was defined in terms of its main objective, operational environment, expected spacecraft configuration, key activities, possible off-nominal conditions, and approximate duration. The complete ConOps baseline is reported in **Error! Reference source not found.**

The baseline ConOps was selected to maximise the scientific return while maintaining a controlled end-of-life strategy. Science operations, orbit maintenance, eclipse survival, and disposal are treated as distinct mission functions, preventing the major Earth eclipse from driving the nominal science-mode design. The spacecraft therefore complete the nominal science phase before transitioning to a reduced-power, disposal-oriented configuration, avoiding excessive power and thermal margins for a single worst-case event occurring late in the mission.

Although the minimum mission-duration requirement could have been satisfied with a shorter operational phase, the baseline adopts a 720-day nominal science phase. This duration increases the radiation-monitoring data volume, improves the statistical robustness of the biological exposure experiment, and enables sampling over a wider range of orbital positions, illumination conditions, and temporal variations. It also reduces the impact of temporary outages, communication gaps, safe-mode events, payload interruptions, or individual spacecraft degradation on the final dataset.

The extended duration is supported by the selected trajectory strategy. The quasi-frozen operational orbit and the exploitation of natural orbital evolution limit the station-keeping demand, while the event-based periselenium control logic avoids unnecessary manoeuvres. As a result, station keeping remains a secondary contributor to the total Delta-V budget, preserving propellant margin for the disposal sequence. The additional operational duration is therefore considered acceptable from a propulsive standpoint.

The 720-day phase also improves the exploitation of the four-spacecraft architecture. A longer science phase increases the benefit of distributed measurements, redundancy, and measurement continuity, while reducing the mission-level impact of partial data losses or single-spacecraft performance degradation. Since the main launch, transfer, deployment, commissioning, and orbit-acquisition costs are already incurred before science operations, extending the operational phase improves the ratio between scientific return and fixed mission cost.

Alternative operational concepts were assessed during preliminary design. Maintaining nominal science operations through the major Earth eclipse would have increased subsystem sizing margins without providing proportional scientific benefit. Conversely, ending the science phase before eclipse entry and transitioning to a reduced-power configuration provides a safer and more efficient mission timeline. The selected ConOps therefore accepts higher operations duration and cumulative exposure to anomalies in exchange for increased scientific return, improved dataset robustness, better use of the four-spacecraft architecture, and preserved end-of-life compliance.

Table 9-2: Baseline Concept of Operations

Sub-phase	Characteristics	Description	Off Nominals	Duration
Pre-launch and Launch Readiness	Objective: To verify launch, space segment, and ground segment readiness. Environment: Ground / launch site.	This phase includes final spacecraft integration, interface verification with the mothercraft, launch vehicle integration, ground segment validation, and final operational readiness review.	Ground segment delay, launch vehicle integration anomaly, spacecraft late non-conformance.	1-2 weeks
Launch and Early Operation	Objective: To deliver the mothercraft and hosted spacecraft to the required departure conditions. Environment: Launch and early Earth-orbit operations.	This phase includes launch, ascent, initial spacecraft/mothercraft activation, telemetry acquisition, and verification of the hosted spacecraft interfaces after launch.	Launcher underperformance, injection error, early telemetry loss, interface anomaly.	4-7 days
Lunar Transfer	Objective: To transfer the mothercraft and hosted spacecraft to the lunar environment. Environment: Earth–Moon transfer trajectory.	This phase includes lunar transfer preparation, transfer injection, cruise operations, trajectory monitoring, and preparation for lunar orbit insertion. The hosted spacecraft remain in a safe configuration with limited health monitoring.	Transfer trajectory dispersion, communication anomaly, thermal or power constraint.	3-5 days
Lunar Orbit Insertion and Deployment	Objective: To insert the mothercraft into lunar orbit and release the four spacecraft. Environment: Initial lunar orbit.	This phase includes lunar orbit insertion, deployment sequence preparation, release of the four spacecraft, detumbling, initial acquisition, and first communication with the mothercraft and/or ground segment.	Deployment failure, excessive angular rates, delayed acquisition, initial orbit dispersion.	2-3 weeks
Spacecraft Commissioning	Objective: To verify spacecraft readiness for orbit acquisition. Environment: Initial lunar orbit.	This phase includes platform checkout, payload functional verification, communication link validation, AOCS calibration, navigation check, power and thermal assessment, and propulsion system readiness verification.	Subsystem anomaly, propulsion unavailability, attitude-control anomaly, telemetry loss.	1-2 weeks
Orbit Acquisition	Objective: To transfer each spacecraft to the operational quasi-frozen lunar orbit. Environment: Lunar transfer orbit / operational orbit insertion.	This phase includes finite low-thrust manoeuvres from the initial lunar orbit to the 100 km × 1500 km operational quasi-frozen orbit, followed by orbit stabilisation and attitude reconfiguration for nominal operations.	Manoeuvre underperformance, orbit determination error, excessive deviation from target orbital elements.	79 days
Science Payload Activation	Objective: To activate and validate the biological and radiation-monitoring payloads. Environment: Operational lunar orbit.	This phase includes payload activation, initial data acquisition, functional checks, and verification of payload compatibility with the operational environment before the start of routine science operations.	Payload activation anomaly, data acquisition error, thermal or power non-compliance.	1-3 days
Science Operations and Orbit Maintenance	Objective: To collect radiation and biological experiment data while maintaining the operational orbit. Environment: Quasi-frozen lunar orbit.	This phase includes routine payload operations, data storage and downlink, spacecraft health monitoring, orbit determination, and station-keeping manoeuvres triggered by the periselenium control logic.	Payload anomaly, data storage saturation, communication outage, excessive orbital drift, station-keeping execution error.	720 days
Reduced-Power Eclipse Configuration	Objective: To survive the major Earth eclipse outside nominal science operations. Environment: Lunar orbit / disposal preparation.	This phase includes transition to a reduced-power configuration, deactivation of non-essential loads, survival through the eclipse, and post-eclipse recovery before continuation of end-of-life operations.	Battery depletion, thermal excursion, incomplete recovery, delayed communication reacquisition.	1 day
End-of-Life Disposal and Passivation	Objective: To remove each spacecraft from the operational orbit and place it in a safe final configuration. Environment: Disposal trajectory / graveyard orbit.	This phase includes disposal manoeuvre execution, final orbit targeting, confirmation of disposal state, deactivation of non-essential functions, and spacecraft passivation as applicable.	Insufficient residual propellant, incomplete disposal, attitude or propulsion anomaly, loss of final telemetry.	120 days + end of life

During the definition of the ConOps, in accordance with the MOs and MDs, a nadir-pointing strategy for the LunaRAY payload was identified as a key mission design choice to ensure consistent characterization of the lunar radiation environment.

The lunar radiation environment comprises GCR, SPE, and secondary albedo particles generated in the lunar regolith. While GCR are approximately isotropic and SPE are directionally dependent on solar activity, albedo particles are strongly influenced by surface interactions and viewing geometry.

A continuous nadir-pointing attitude throughout the orbit enables high-fidelity measurements of lunar albedo at periselenium. It provides a partially shielded viewing configuration at aposelenium, supporting the reconstruction of radiation conditions representative of the lunar surface environment.

9.3 Functional Analysis

The Functional Analysis is a structured and systematic decomposition process aimed at identifying, organizing, and allocating the functions that the system shall perform to fulfil the mission objectives. This process enables the traceability between high-level mission goals and the functions implemented at subsystem level across the entire system lifecycle.

In early design phases, Functional Analysis constitutes a key activity for the definition of functional architecture and the derivation of functional requirements. It provides a solution-independent description of the system's behaviour, supporting consistency, completeness, and interface identification.

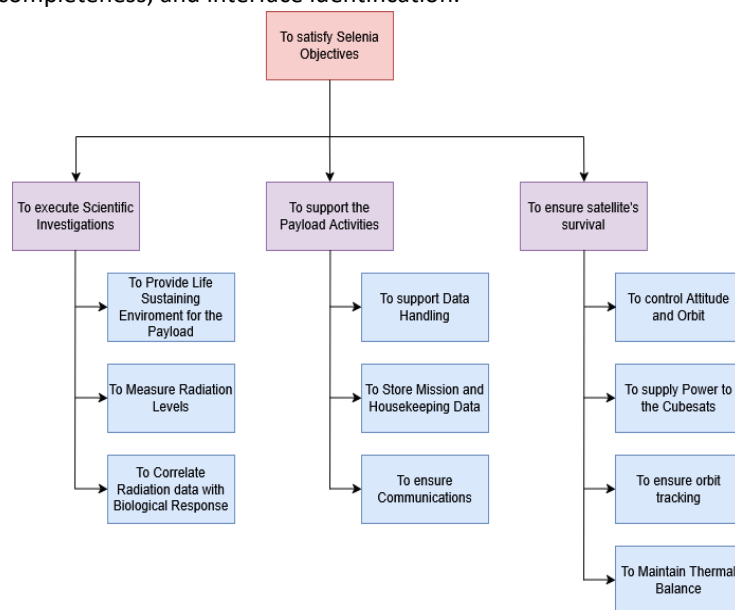


Figure 9-1: High Level Functional Tree

For the SELENIA mission, the Functional Tree has been developed up to three levels of decomposition. At the highest level, a limited number of primary functions are identified, representing the main mission capabilities. These are further decomposed into subfunctions and detailed functional tasks. The first level includes three main functions, as shown in Figure 9-1. These correspond to: executing scientific investigations, supporting payload activities, and ensuring satellite survival. Each of them is further divided into subfunctions and lower-level functional tasks. For example, the function “to execute scientific investigations” includes providing a life-sustaining environment for the payload, measuring radiation levels, and correlating radiation data with biological response.

9.4 Mission Architecture(s)

The mission architecture was defined through a trade-off between alternative space-segment and orbit-geometry solutions. The objective of the trade-off was to identify the architecture providing the best compromise between operational availability, safety and redundancy, radiation-exposure prediction capability, and fuel consumption.

Four candidate architectures were assessed. The first option considered a single spacecraft on a highly elliptical lunar orbit. The second option adopted four spacecraft on the same highly elliptical lunar orbit. The third option considered a single spacecraft on a low frozen lunar orbit, while the fourth option adopted four spacecraft on a low frozen lunar orbit. The architectural trade-off was performed through a weighted multi-criteria decision matrix, following an AHP-like approach. Each candidate architecture was evaluated against the selected drivers, and a final score was obtained as the weighted sum of the normalized scores assigned to each criterion. The high-elliptical orbit options provide a more favourable fuel-consumption performance and improved access

conditions, while the low frozen orbit options improve the prediction of radiation exposure effects due to the more stable and repetitive orbital geometry. The single-spacecraft options reduce operational complexity but introduce a single-point failure at mission level. Conversely, the four-spacecraft configurations increase mission robustness and reduce the impact of individual spacecraft degradation or loss.

Configuration	Low Outage Probability		Safety and Redundancy		Prediction of Radiation Exposure Effects		Optimize Fuel Consumption		Final Score
	Weighting	Score	Weighting	Score	Weighting	Score	Weighting	Score	
Configuration concept (number of satellites in-plane + inclination type)									
Configuration 1 — 1 + High Elliptical	0.232	0.157	0.232	0.157	0.102	0.167	0.196	0.375	0.164
Configuration 2 — 4 + High Elliptical	0.232	0.398	0.232	0.398	0.102	0.167	0.196	0.375	0.275 <i>Selected baseline</i>
Configuration 3 — 1 + Low Frozen	0.232	0.130	0.232	0.130	0.102	0.333	0.196	0.125	0.119
Configuration 4 — 4 + Low Frozen	0.232	0.315	0.232	0.315	0.102	0.333	0.196	0.125	0.205

Figure 9-2: Weighted Multi-Criteria Evaluation of Candidate Orbital Configurations

The selected baseline is the four-spacecraft architecture on a highly elliptical lunar orbit. This solution provides the best overall compromise between outage probability, redundancy, and fuel consumption, while maintaining sufficient capability for radiation-environment sampling and biological exposure monitoring. The four-spacecraft configuration improves mission resilience and measurement availability, whereas the common orbit geometry allows standardisation of deployment, orbit acquisition, station keeping, and disposal operations across the space segment.

The baseline architecture therefore comprises four identical spacecraft released into lunar orbit by a mothercraft, with each spacecraft phased in the operational orbit by a 5° true anomaly separation. Each spacecraft is transferred to the selected operational orbit and operated according to the same nominal mission logic. Differences between spacecraft are limited to deployment timing, phasing conditions, manoeuvre execution epochs, and communication-window availability. This approach reduces the complexity associated with multi-orbit architectures while preserving the benefits of a distributed space segment.

9.5 Mission Analysis

The mission analysis verifies the feasibility of the selected architecture from an orbital and propulsive standpoint. The analysis covers the trajectory geometry, the finite low-thrust manoeuvre sequence, the operational orbit maintenance strategy, the constellation phasing logic, and the end-of-life disposal trajectory.

The reference case is defined at single-spacecraft level and then applied to the four-spacecraft space segment. This approach is consistent with the selected architecture, since all spacecraft share the same operational orbit and manoeuvre logic, while individual manoeuvre epochs may differ according to deployment timing, phasing requirements, and orbit-determination updates.

The following subsections report the trajectory design and the resulting Delta-V budget. Detailed orbit parameters, manoeuvre timing, and Delta-V contributions are provided only where they directly support the baseline mission design.

9.5.1 Trajectory Design

The trajectory design is based on an initial lunar orbit provided by the mothercraft and a final high-elliptical quasi-frozen operational orbit with nominal periselenium and aposelenium altitudes of 100 km and 1500 km, respectively.

The transfer strategy exploits the natural periselenium evolution of the initial lunar orbit. The initial argument of periselenium is selected so that lunar gravitational perturbations and solar radiation pressure drive the periselenium

altitude toward the target value. This reduces the deterministic orbit-acquisition cost by avoiding a dedicated periselenium-lowering manoeuvre.

Orbit acquisition is completed through finite low-thrust manoeuvres targeting the final operational orbit. The finite-burn implementation is retained in the mission analysis to ensure consistency with the selected propulsion system, including thrust level, burn duration, and power demand. The manoeuvre sequence is designed to control the targeted orbital elements while limiting deviations in non-targeted elements.

The four-spacecraft geometry is established by phasing the spacecraft along the same operational orbit. The use of a common reference orbit avoids the additional Delta-V and operational complexity associated with multi-orbit or multi-plane configurations. Phasing is achieved through deployment timing, drift duration, and manoeuvre-epoch selection, while maintaining safe separation during orbit acquisition and nominal operations.

During the science phase, orbit maintenance is based only on periselenium-altitude control. A station-keeping manoeuvre is triggered when the periselenium altitude deviates from the nominal value by more than ± 5 km. This event-based strategy limits unnecessary manoeuvres and constrains the cumulative maintenance Delta-V.

The disposal trajectory starts after completion of the nominal science phase. The sequence consists of an aposelenium-raising phase followed by an eccentricity-reduction phase targeting a nearly circular 5000 km $\times$ 5000 km disposal orbit. This orbit was selected to provide a stable end-of-life configuration sufficiently separated from the operational orbit, thereby reducing the probability of long-term interference with the science trajectory. The nearly circular geometry also limits the recurrence of low-altitude passages over the lunar surface and provides a more predictable post-disposal evolution, supporting safe spacecraft passivation and compliance with end-of-life mission constraints.

9.5.2 DeltaV Budget

The Delta-V budget was derived from the finite low-thrust manoeuvre summary generated for the reference spacecraft. The budget reported in Table 9-3 refers to a single spacecraft, since each spacecraft is equipped with its own propulsion system and executes the same nominal manoeuvre sequence. For the four-spacecraft architecture, the cumulative space-segment equivalent is obtained by scaling the single-spacecraft budget by the number of spacecrafts; this value is used only for mission-level accounting and does not represent the Delta-V capability required by an individual spacecraft.

The manoeuvre budget includes orbit acquisition, station keeping, and end-of-life disposal. Each manoeuvre group is reported separately according to its function in the trajectory design. The deterministic single-spacecraft Delta-V is 903.58 m/s, with a total propellant consumption of 1.732 kg. A 10% design margin is applied to account for navigation uncertainties, thrust dispersion, manoeuvre execution errors, orbit-determination inaccuracies, and contingency corrections. The resulting margined Delta-V requirement is 993.94 m/s per spacecraft.

Table 9-3: Single-spacecraft deterministic Delta-V budget

Manoeuvre Group	Mission Phase	Number of Finite Burns	Deterministic Delta-V [m/s]	Delta-V + 10% Margin [m/s]	Propellant Used [kg]	Purpose
Target aposelenium to 1500 km	Orbit acquisition	351	321.19	353.31	0.616	Acquisition of the 100 km $\times$ 1500 km operational orbit
Station Keeping	Science Operations	126	109.62	120.58	0.210	Maintenance of the operational orbit during the science phase
Target aposelenium to 5000 km	Disposal	286	237.41	261.15	0.455	First disposal phase, raising the aposelenium toward the disposal orbit
Target eccentricity to zero	Disposal	157	235.36	258.89	0.451	Final disposal phase, targeting a nearly circular 5000 km $\times$ 5000 km orbit
Total	Full Mission	920	903.58	993.94	1.732	Total single-spacecraft budget

The disposal sequence is the dominant contributor to the deterministic Delta-V budget, with a combined contribution of 472.77 m/s. Orbit acquisition requires 321.19 m/s, while station keeping remains limited to 109.62 m/s over the analysed science phase. This distribution confirms that the selected operational orbit limits the maintenance cost, whereas end-of-life disposal remains the main propulsion driver.

At constellation level, the deterministic cumulative Delta-V equivalent is 3614.31 m/s. Including the 10% margin, the cumulative space-segment equivalent becomes 3975.74 m/s. Propulsion sizing is performed on the single-spacecraft margined requirement of 993.94 m/s.

9.6 Risk Assessment

Risk analysis is a very important tool because it allows us to identify, classify, and reduce both project and mission risk, ensuring a higher mission success rate. It is important that the risk analysis is performed from the early stages of the project, as it should act as an input for mission design.

A detailed explanation of the risk identification and mitigation methods is provided in Chapter 12.5.

A dedicated risk register was compiled for every identified risk scenario. Additionally, a fault tree was developed for technical risks when required. Here are two examples:

Table 9-4: Risk Registers

Risk Register					
Project:	Selenia		Source:	Federico Sancandi	Selenia
WBS Ref:	WP02.1		Assigned to:	Federico Sancandi	Selenia
Date:	24/04/2026		Controlled by:		
Issue:	1.0		Approved by:		
Risk Scenario and Decision					
Risk No.	TCH-04	Risk scenario title:	Inability to collect desired mission data		
Cause:	- Inability of the dehydrated biological samples to successfully reactivate upon rehydration (eukaryotes exhibit the highest vulnerability to this failure) - Death of the biological payload due to a TCS fault * - Loss of attitude control * - Loss of OBDH capabilities * - Loss of communication with the orbiter * - UART/RT- 485 transceivers fail to transfer payload data to the OBC - Memory overflow due to missed communication windows				
Consequence:	Failure to achieve the primary scientific objective				
Likelihood (L)	Severity (S)	Risk Index (before)	Risk domain	Risk Decision	Status
High	Critical	D4	Technical	Reduce	Open with action
Risk Mitigation					
Risk reduction measure:	- Reduce satellite orbit transfer time to limit degradation of the dehydrated species - Implement a Survival Mode that protects payload and spacecraft integrity during anomaly or fault conditions - Add a fourth satellite as backup to provide redundancy - Install an external backup thermal sensor near the biological payload to monitor survival temperature limits - Size the onboard memory capacity to store up to one week of data in the event of a communication loss with the orbiter				
Verification means:	Review of design, Analysis, Test				
Expected risk reduction:	Likelihood (L)	Severity (S)	Risk Index (after)		
	Low	Critical	B4		
Action(s):	- Define the FDIR logic and thresholds required to trigger the automatic Survival Mode - Implement a store-and-forward protocol on the OBC - Update the architecture block diagram and the mass/power budgets to account for the added sub-systems				

Risk Register					
Project:	Selenia		Source:	Federico Sancandi	Selenia
WBS Ref:	WP04.2		Assigned to:	Federico Sancandi	Selenia
Date:	24/04/2026		Controlled by:		
Issue:	1.0		Approved by:		
Risk Scenario and Decision					
Risk No.	TCH-08	Risk scenario title:	Loss of attitude control (ADCS fault)		
Cause:	- Loss of electrical power * - Loss of OBDH capabilities * - Inability to desaturate reaction wheels due to reaction control thrusters fault - Mechanical jamming of a reaction wheels - Reaction wheel overheating and mechanical degradation caused by excessive bearing friction - IMU malfunction, excessive drift, or data corruption - Loss of valid attitude data due to sun sensor degradation, malfunction or eclipse - Single event upset in the GNC/ADCS process unit - Single event latch-up in the GNC/ADCS process unit - GNC/ADCS control unit hardware fault - ADCS software anomaly				
Consequence:	Inability to maintain spacecraft pointing, leading to payload operations interruption and inability to ensure stationkeeping				
Likelihood (L)	Severity (S)	Risk Index (before)	Risk domain	Risk Decision	Status
Medium	Critical	C4	Technical	Reduce	Open with action
Risk Mitigation					
Risk reduction measure:	- Use 8 reaction control thrusters and implement an autonomous momentum management software to allow thruster-based desaturation - Implement a 4-reaction wheel configuration to provide redundancy and single-fault tolerance - Size the reaction wheels with conservative momentum storage margins - Select high-reliability reaction wheels and thrusters with a qualified lifetime that significantly exceeds the mission duration, to prevent premature wear-out - Use 2 IMU to provide hardware redundancy - Implement sensor redundancy by using 3 star tracker and 8 sun sensors (2 Fine Sun Sensors, 6 Coarse Sun Sensors as backup) to heavily reduce the risk of losing attitude data - Implement error correction code memory to mitigate single event upsets (SEU) - Implement latch-up current limiters to mitigate the effects of single event latch-ups (SEL) - Implement an automatic Safe/Survival Mode triggered by critical anomalies or ground commands, designed to continuously transmit raw and processed sensor telemetry to the OBC for ground-based troubleshooting if needed				
Verification means:	Review of Design, Test				
Expected risk reduction:	Likelihood (L)	Severity (S)	Risk Index (after)		
	Minimum	Critical	A4		
Action(s):	- Update the ADCS architecture block diagram and the mass/power budgets to account for the added sub-systems - Develop the control algorithms for the autonomous thruster-based momentum management software - Define the FDIR logic and thresholds required to trigger the automatic Safe/Survival Mode				

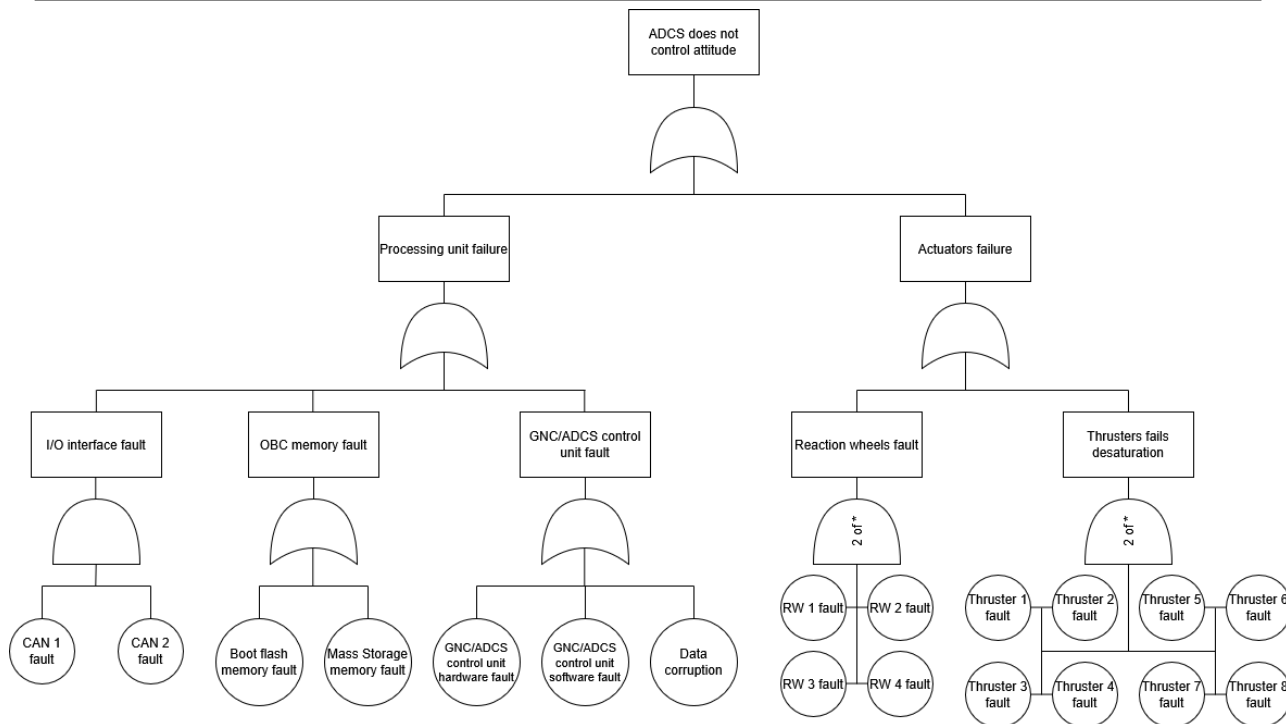


Figure 9-3: Fault Tree

Note: The asterisk (*) represents a combination logic, meaning that a certain number of components must fail for the upper-level fault to occur. This was done to avoid defining all the different failure combinations, which would make the Fault Tree too large and difficult to read.

All risk registers are summarized in the following risk log:

Table 9-5: Risk Log

ID	Risk Scenario	Cause	Risk index (before)	Risk reduction measure	Risk index (after)
C-01	Inaccurate cost margin definition	- Dramatic changes in global economy - Inability to correctly predict hidden costs that might arise during mission development - Optimism bias	B3	Considering an independent cost estimation conducted by external reviewers as a primary tool to eliminate optimism bias and ensure adequate funding is allocated for the risk mitigation and validation phases	A3
SCH-01	Missed delivery of the phase A final review package	- Lack of active participation of key team members during critical design phases - Inaccurate estimation of the time required for subsystem-level design - Inadequate communication and information flow between team members - Inefficient teamwork - Failure to adequately define and baseline mission requirements and deliverables	C3	- Develop a detailed work logic and establish a baseline schedule with sufficient time margins for critical activities - Enforce weekly progress tracking meetings - Enforce mandatory attendance for concurrent design sessions to ensure continuous subsystem alignment - Ensure that all project changes and design updates are promptly communicated to every team member	A3
SCH-02	Space segment isn't ready to launch for 2030	- Failure to complete mission design in time - Unforeseen challenges during manufacturing - Repeated failure during qualification testing	C4	- Strictly using off-the-shelf components to drastically reduce custom manufacturing and R&D time - Complete major programmatic milestones strictly on their baseline dates - Prioritize the early purchase of critical hardware that requires extended manufacturing times	B4
TCH-01	Space segment fails to meet launch and integration requirements	- The space segment mass exceeds 100kg - The space segment volume exceeds 12U - Damages during transport - Failing final readiness review	C4	- Allocate sufficient mass and volume margins to account for potential design evolution and component growth during development - Use proper packaging and safe handling procedures during transport	B4
TCH-02	Satellites are unable to safely reach the Moon	- Launcher separation failure - Launcher fails to reach space - Space debris impact - Satellites are unable to survive launch loads due to inadequate structural design - Satellites are unable to survive launch loads due to manufacturing defects - Satellites are unable to survive launch loads due to dynamic coupling with the launcher's fundamental frequencies - Collision with orbiter during deployment - Inter-satellite collision during deployment	B5	- Select a highly reliable launch vehicle with a proven track record for lunar insertions - Design the satellite structure to withstand the maximum expected static and dynamic launch loads - Ensure the satellite's fundamental frequencies are decoupled from the launcher's natural frequencies - Implement strict quality assurance protocols and environmental screening - Design a safe deployment sequence to ensure adequate physical clearance between the orbiter and the satellites	A5

TCH-03	Satellites are unable to enter and keep desired orbit	- Payloads are unable to correctly detach from orbiter due to a release mechanism malfunction, which results in asymmetric deployment or partial separation - Communication loss induced by severe spacecraft tumbling after deployment - Orbiter fails to release satellites in correct orbit - Loss of propulsion system * - Loss of attitude control * - Landmark tracking camera fault (optical, electrical, or software anomaly) - GNC/ADCS control unit hardware fault - GNC software anomaly - Higher than expected propellant consumption (due to thrusters underperformances, micro-leaks in the propellant feed system or unmodeled perturbations)	C4	- Implement a Detumbling Mode to recover from asymmetric deployment or abnormal rotation rates during deployment - Enable simultaneous communication through both LGAs to prevent loss of communication while tumbling - Use flight-proven separation mechanisms equipped with redundant release triggers - Allocate a delta-V margin to correct small orbit injection errors - Implement IMU-based Dead Reckoning to propagate spacecraft state during temporary landmark camera faults - Implement a Safe Navigation Mode that relies only on Sun Sensors and state propagation, to save power in Safe/Survival Mode	B4
TCH-04	Inability to collect desired mission data	- Inability of the dehydrated biological samples to successfully reactivate upon rehydration (eukaryotes exhibit the highest vulnerability to this failure) - Death of the biological payload due to a TCS fault * - Loss of attitude control * - Loss of OBDH capabilities * - Loss of communication with the orbiter * - UART/RS-485 transceivers fail to transfer payload data to the OBC - Memory overflow due to missed communication windows	D4	- Reduce satellite orbit transfer time to limit degradation of the dehydrated species - Implement a Survival Mode that protects payload and spacecraft integrity during anomaly or fault conditions - Add a fourth satellite as backup to provide redundancy - Install an external backup thermal sensor near the biological payload to monitor survival temperature limits - Size the onboard memory capacity to store up to one week of data in the event of a communication loss with the orbiter	B4
TCH-05	Inability to collect housekeeping data	- Loss of OBDH capabilities * - Loss of communication with the orbiter * - Housekeeping sensor anomalies - Platform CAN bus fault - CAN transceivers fail to transfer housekeeping data to the OBC - Memory overflow due to missed communication windows	B4	- Use radiation tolerant sensors - Add a redundant platform CAN bus to ensure continuous internal communication - Size the onboard memory capacity to store up to one week of data in the event of a communication loss with the orbiter - Add redundant telemetry sensors (for temperature, power, and status) for all critical subsystems	A4
TCH-06	Satellites are unable to keep desired orbit for the full mission duration	- Loss of propulsion system * - Loss of attitude control * - Loss of communication with the orbiter * - Space debris impact - Landmark tracking camera fault (optical, electrical, or software anomaly) - GNC/ADCS control unit hardware fault - GNC software anomaly - Higher than expected propellant consumption (due to thrusters underperformances, micro-leaks in the propellant feed system or unmodeled perturbations) - Thruster misalignment causing parasitic torques during orbital station-keeping - Loss of attitude control * - Loss of communication with the orbiter * - Loss of propulsion system * - Higher than expected propellant consumption (due to thrusters underperformances, micro-leaks in the propellant feed system or unmodeled perturbations) - Landmark tracking camera fault (optical, electrical, or software anomaly) or adverse lunar illumination conditions - GNC/ADCS control unit hardware failure or software anomaly during the disposal maneuver calculation or execution	C3	- Allocate a delta-V margin to account for unmodeled lunar gravity perturbations - Select a radiation resistant coverglass for the solar array, and oversize the arrays to guarantee adequate end of life power generation - Use radiation tolerant electronic components - Use highly reliable battery with as little depth of discharge as possible - Implement a Safe Navigation Mode that relies only on Sun Sensors and state propagation, to save power in Safe/Survival Mode - Implement IMU-based Dead Reckoning to propagate spacecraft state during temporary landmark camera faults	B3
TCH-07	Inability to safely decommission the cubesats	- Loss of attitude control * - Loss of propulsion system * - Higher than expected propellant consumption (due to thrusters underperformances, micro-leaks in the propellant feed system or unmodeled perturbations) - Landmark tracking camera fault (optical, electrical, or software anomaly) or adverse lunar illumination conditions - GNC/ADCS control unit hardware failure or software anomaly during the disposal maneuver calculation or execution	C2	- Allocate a delta-V margin to account for unmodeled lunar gravity perturbations - Implementing an autonomous software failsafe that automatically executes a safe disposal maneuver upon detecting an unrecoverable loss of communication - Implement IMU-based Dead Reckoning to propagate spacecraft state during temporary landmark camera faults - Adding an hard-coded propellant reserve dedicated exclusively to disposal maneuvers	B2
TCH-08	Loss of attitude control (ADCS fault)	- Loss of electrical power * - Loss of OBDH capabilities * - Inability to desaturate reaction wheels due to reaction control thrusters fault - Mechanical jamming of a reaction wheels - Reaction wheel overheating and mechanical degradation caused by excessive bearing friction - IMU malfunction, excessive drift, or data corruption - Loss of valid attitude data due to sun sensor degradation, malfunction or eclipse - Single event upset in the GNC/ADCS process unit - Single event latch-up in the GNC/ADCS process unit - GNC/ADCS control unit hardware fault - ADCS software anomaly	C4	- Use 8 reaction control thrusters and implement an autonomous momentum management software to allow thruster-based desaturation - Implement a 4-reaction wheel configuration to provide redundancy and single-fault tolerance - Size the reaction wheels with conservative momentum storage margins - Select high-reliability reaction wheels and thrusters with a qualified lifetime that significantly exceeds the mission duration, to prevent premature wear-out - Use 2 IMU to provide hardware redundancy - Implement sensor redundancy by using 3 star tracker and 8 sun sensors (2 Fine Sun Sensors, 6 Coarse Sun Sensors as backup) to heavily reduce the risk of losing attitude data - Implement error correction code memory to mitigate single event upsets (SEU) - Implement latch-up current limiters to mitigate the effects of single event latch-ups (SEL) - Implement an automatic Safe/Survival Mode triggered by critical anomalies or ground commands, designed to continuously transmit raw and processed sensor telemetry to the OBC for ground-based troubleshooting if needed	A4
TCH-09	Loss electrical power (EPS fault)	- Solar array degradation due to long term usage, radiation exposure or thruster plume impingement - Solar array deployment mechanism failure (SADA failure) - Inability to maintain sun-pointing tracking due to solar array drive mechanism failure or total loss of attitude control - Power generation shortfall caused by shadows cast from deployed antenna - Failure of one or more battery packs - Power bus damage due to single event latch-up (SEL) - Cable damage/short circuits caused by mechanical stress or thermal cycling (PCDU fault) - Thermal control system fails to keep battery temperature in the operational limits - Battery thermal runaway	C5	- Oversize the solar arrays to provide sufficient margin against potential power losses, due to radiation degradation and aging - Add radiation resistant coverglass for the solar array - Use radiation tolerant electronic components - Select highly reliable batteries with as little depth of discharge as possible - Implement latch-up current limiters (LCL) in the regulated power bus to mitigate the effects of single event latch-ups (SEL) - Implement a three pack battery architecture to provide redundancy and tolerate a single pack failure - Implement an autonomous Safe/Survival Mode that automatically acquires and maintains a Sun-pointing attitude whenever the battery state of charge drops below 20%, ensuring solar array recharging - Add backup thermal sensors near the battery packs	A5

TCH-10	Loss of thermal control (TCS fault)	- Loss of electrical power * - Loss of OBDH capabilities * - Degradation or physical damage to the heaters due to radiation - Degradation or physical damage to the multi-layer insulation (MLI) and thermal coatings - Degradation or physical damage to the thermal sensors - Prolonged exposure to abnormally long eclipse periods - Off-nominal sun pointing - Loss of attitude control * - Insufficient heat dissipation capacity during thruster firing phases - TCS software failure	C4	- Add redundant thermal sensors across all critical subsystems and the payload to ensure comprehensive temperature monitoring - Add 27W heaters to prevent thermal drops below operational thresholds during orbital eclipses - Add thermal measures to handle heat during thruster use, such as deployable radiators or thruster isolation - Ensure solar array cooling through an high emissivity rear coating - Optimize the internal layout to ensure sensitive components are safely isolated from major heat sources	B4
TCH-11	Loss of propulsion system (Propulsion system fault)	- Loss of electrical power * - Loss of OBDH capabilities * - Main thruster fault (neutralizer cathode failure, internal short circuits or premature wear-out) - Power processing unit (PPU) fault - Propellant leak in the feed system - Propellant leak in the tanks - Dramatic increase in tank pressure - Feeding valve fails to correctly open - Propellant backflow - Propellant impurity clogs the feeding lines	C4	- Add a check valve to prevent backflow - Add filters to prevent valve clogging due to impurities - Add a pressure sensor inside the tank to monitor pressure - Add pressure relief valve to prevent possible rupture of the pressurized tanks	B4
TCH-12	Loss of communication with the orbiter (TT&C fault)	- Catastrophic loss of attitude control * - Loss of electrical power * - Loss of OBDH capabilities * - Satellites are unable to communicate with orbiter due to LGA malfunctions - Satellites are unable to communicate with orbiter due to LGA deployment mechanism malfunction - Mechanical interference during deployment leading to mutual hardware damage - UHF band transceiver fault	D5	- Implement a redundant antenna deployment mechanism - Use radiation-tolerant electronic components - Implement a dual LGA architecture to ensure full spherical coverage during the detumbling phase and ensure single-fault tolerance	A5
TCH-13	Loss of on-board data handling capabilities (OBC fault)	- Loss of electrical power * - Loss of the input/output interface (CAN/UART/RS-485 transceiver malfunction) - Data corruption or system crash/corruption due to single event upset (SEU) - Hardware damage due to single event latch-up (SEL) - Software bugs and logic errors - On-board memory overflow due to missed communication windows - TCS fails to maintain the OBC within its safe operational temperature limits *	C5	- Use radiation tolerant electronic components - Split the memory storage into Mass Storage and Boot Image Flash to ensure a multi-layer memory protection, to ensure a safe reboot procedure - Size the onboard memory capacity to store up to one week of data in the event of a communication loss with the orbiter - Implement error correction code memory to mitigate single event upset (SEU) - Implement latch-up current limiters (LCL) in the OBC to mitigate the effects of single event latch-ups (SEL) - Add backup thermal sensors near the MCU to ensure temperature monitoring - Add an external watchdog to automatically reset the OBC in the event of a software bugs or processor freeze	A5

Note: The risk scenarios highlighted in green are technically root causes of higher-level risks. However, we've treated them as independent risk scenarios because they act as common points of failure, making it easier to analyse and mitigate them across the whole project.

Note: Causes marked with an asterisk (*) represent separate risk scenarios that are fully analysed and mitigated in their respective Risk Register.

As shown in the risk registers, specific mitigation measures were proposed for each risk to lower its risk index. The following figures show the risk matrices both before and after the implementation of these risk reduction strategies:

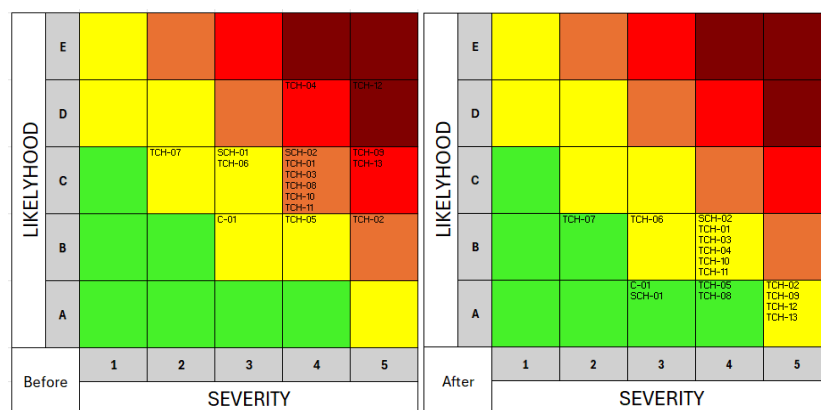


Figure 9-4: Risk Matrices before and after Risk Reduction

As shown in the post-mitigation matrices, all identified risks have been successfully reduced to acceptable levels, primarily through a fault-tolerant design philosophy. This result was achieved by allocating adequate mass and volume margins, implementing hardware redundancy, conducting dedicated analyses, and holding periodic meetings to monitor the risk reduction measures. However, certain root causes cannot be fully mitigated as they depend entirely on external factors. These include launch operation failures, launcher or orbiter anomalies, and mission budget overruns tied to global economic trends.

10 System Design

This section presents the SELENIA system design and its subsystems. It includes an overview of system requirements and system architecture and budgets, followed by the definition of the operational modes. Each subsystem is then described in terms of requirements, drivers, architecture, and sizing. Relevant CAD models, layouts, and configuration aspects are also provided. The section supports the subsequent AIV definition.

10.1 System Requirements

Following the definition of the HLR and MR, the system-level requirements were established. For each subsystem, a dedicated set of requirements was developed to guide the subsystem design process. These requirements were derived using different criteria and were defined in accordance with the standards specified by the ECSS regulations. Consequently, they were classified into categories defined by the standard. Each requirement belongs to a specific category, identified in the Type field through an abbreviated code. Apart from the 24 defined **Mission** requirements, the main requirement categories are: **Functional** (39 Requirements), **Interface** (13), **Physical** (14), **Environmental** (10), **Operational** (25), Configuration (7), and **Design** (12). In the end, a total of 120 Requirements has been defined (excluding the Mission ones). Most subsystem requirements were derived either from the Functional Analysis (Functional Requirements) or from the ConOps (Operational Requirements).

Each requirement is assigned a unique identification code. An example of a requirement ID is: **R-EPS-01-02**

In this identifier, EPS denotes the subsystem to which the requirement applies. The primary requirements, defined during the initial stages of the design process and addressing the main aspects of the subsystem, are identified by the first pair of digits (e.g., **R-EPS-01-00**, **R-EPS-02-00**, etc.). More specific requirements, subsequently derived from a primary requirement and linked to it through a parent-child relationship, are identified using the second pair of digits (e.g., while **R-EPS-01-00** is a primary requirement, its “child” requirements are labelled **R-EPS-01-01**, **R-EPS-01-02**, etc). The purpose of this classification scheme is to highlight relationships among requirements and to emphasize the hierarchical structure of the requirements defined for each subsystem.

For each subsystem, only a selection of the most significant requirements, those that had the greatest impact on and imposed the strongest constraints on the design process, will be presented within this report.

Some requirements about the overall system have been defined. Most of them belong to the Interface Category, regarding the relationships and connection between the subsystems. The most determinant ones, for the overall system design, are shown in Table 10-1.

Table 10-1: System Requirements

Requirement ID	Requirement	Type	Verification method
R-SYS-07-00	The system shall implement a main data bus based on the Controller Area Network protocol.	DES	Review, Analysis
R-SYS-07-01	The CAN-based data bus shall support multi-node communication between all connected subsystems.	FUNC	Review, Test
R-SYS-07-02	The system shall provide a communication interface based on UART for the payload.	INTER	Review, Analysis
R-SYS-07-03	Each payload shall maintain a dedicated UART communication link with the OBC for command and telemetry exchange.	INTER	Review, Test

10.2 Operative Modes

The operative modes are defined to describe the configurations of the spacecraft and its subsystems in each phase of the mission based on the specific level of system operativity of each mode. Transitions between modes are governed by well-defined physical conditions, such as angular rates, deployment status, orbital conditions, system health, and mission phase progression. This resulted in the definition of 13 modes organized in a state diagram that defines the spacecraft operational concept as a deterministic state machine, where each mode corresponds to a unique, invariant configuration of subsystem states.

A key design principle underlying this architecture is that each mode must correspond to a single, unambiguous configuration of all subsystems, with no internal variability. This is particularly critical for the payload, whose activation status directly impacts power, thermal, and data handling strategies, and therefore acts as a primary discriminator between operational regimes.

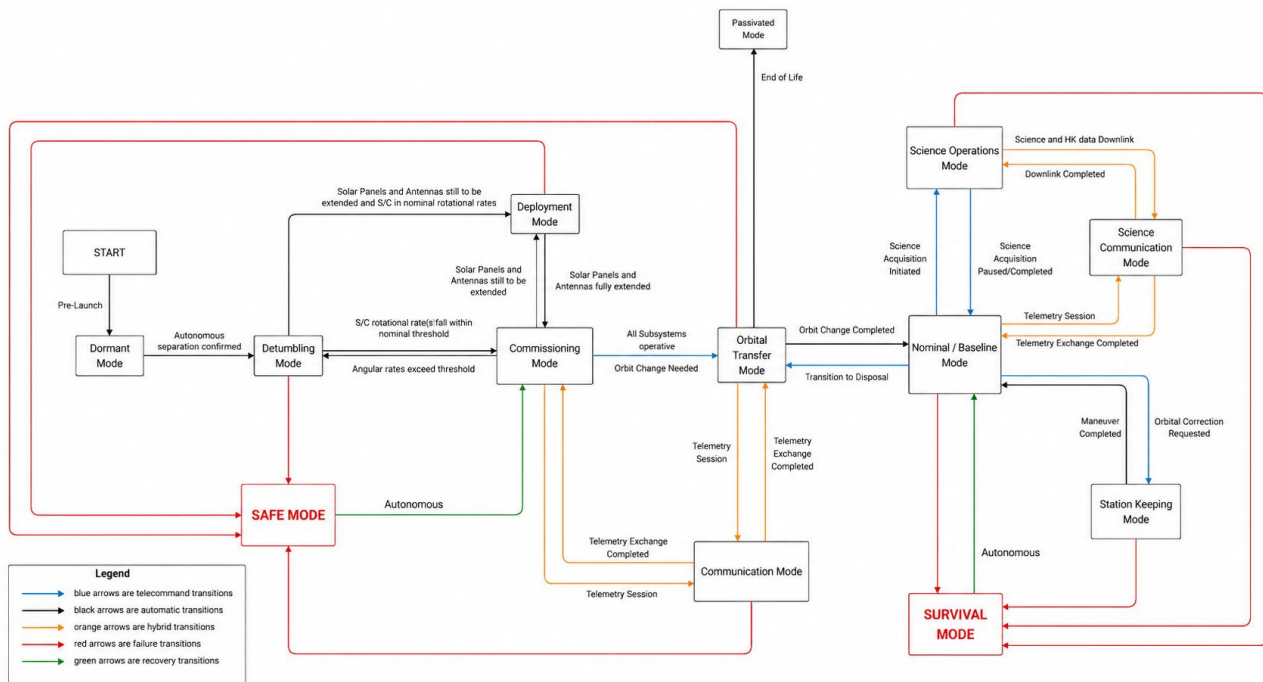


Figure 10-1: State Diagram

The mission starts in **Dormant Mode**, representing the pre-launch configuration. All subsystems are powered OFF, and no onboard autonomy is active. Transition to operational modes occurs only after confirmed separation and availability of onboard power and control.

Following separation, the spacecraft enters **Detumbling Mode**, activated whenever angular rates exceed allowable limits. The system prioritizes attitude stabilization with only essential subsystems active. Exit occurs upon achieving rotational rates compatible with stable control.

Once stabilized, the spacecraft transitions to **Deployment Mode**, during which solar arrays and antennas are deployed. Only subsystems required for mechanical deployment, initial power generation, and deployment verification are active. The spacecraft then enters **Commissioning Mode**, where all platform subsystems (excluding payload) are activated and validated to confirm readiness for operations.

If required, the spacecraft proceeds to **Orbital Transfer Mode**, where propulsion and navigation subsystems are used to reach the target orbit. The payload remains inactive and communication is limited to beacon transmission.

During all pre-payload phases, the spacecraft may enter **Communication Mode** when a communication window is available and either scheduled contact or an external request occurs. In this mode, the communication subsystem supports data exchange within power and thermal constraints.

In case of anomaly, the spacecraft transitions to **Safe Mode**, where non-essential subsystems are powered OFF, operations are reduced to survival-level functions, and power is prioritized for battery charging. Communication operates in a reduced, robust configuration. The payload is powered OFF.

Recovery from Safe Mode leads to **Commissioning Mode** for subsystem revalidation prior to resuming operations.

Upon reaching the operational orbit, the spacecraft enters **Nominal Mode**, a fully powered and stable configuration with all subsystems active. The payload is powered but not yet operational.

If orbital deviations exceed allowable limits, the spacecraft transitions to **Orbit Maintenance Mode**, where propulsion and guidance subsystems perform corrective manoeuvres. Upon stabilization, the system returns to Nominal Mode.

When mission conditions are satisfied, the spacecraft enters **Science Operations Mode**, where the payload is fully operational and drives system activity.

During operational phases, the spacecraft may enter **Science Communication Mode**, functionally equivalent to Communication Mode, but prioritizing payload data downlink under nominal operational constraints.

In case of anomaly after payload activation, the spacecraft transitions to **Survival Mode**, a degraded configuration in which the payload remains powered ON but inactive. Platform and payload are maintained within safe power and thermal limits, and communication supports system monitoring.

Recovery from Survival Mode returns the spacecraft to **Nominal Mode** without requiring full revalidation.

At the end of operations, the spacecraft may re-enter **Orbital Transfer Mode** to perform disposal manoeuvres.

The mission concludes in **Passivated Mode**, where all subsystems are permanently deactivated and residual energy sources are neutralized to ensure compliance with debris mitigation requirements.

10.3 System Architecture

The system architecture connects all spacecraft subsystems through two fundamental infrastructures: the Power Bus and the Data Bus. Together, these two networks form the core of the satellite's internal architecture, ensuring that power and information are efficiently distributed across all subsystems to maintain coordinated and reliable operation. The diagram below provides a clear visual representation of this architecture, showing how each subsystem is interconnected through dedicated power and data links. The color-coded layout highlights the separation between the main Data Bus, the Payload Data Bus, and the regulated Power Bus, where red arrows denotes power exchange, green arrows represents the data link, and orange arrows indicates the payload data link, making it easier to understand how the spacecraft maintains coordination and reliability across all its components.

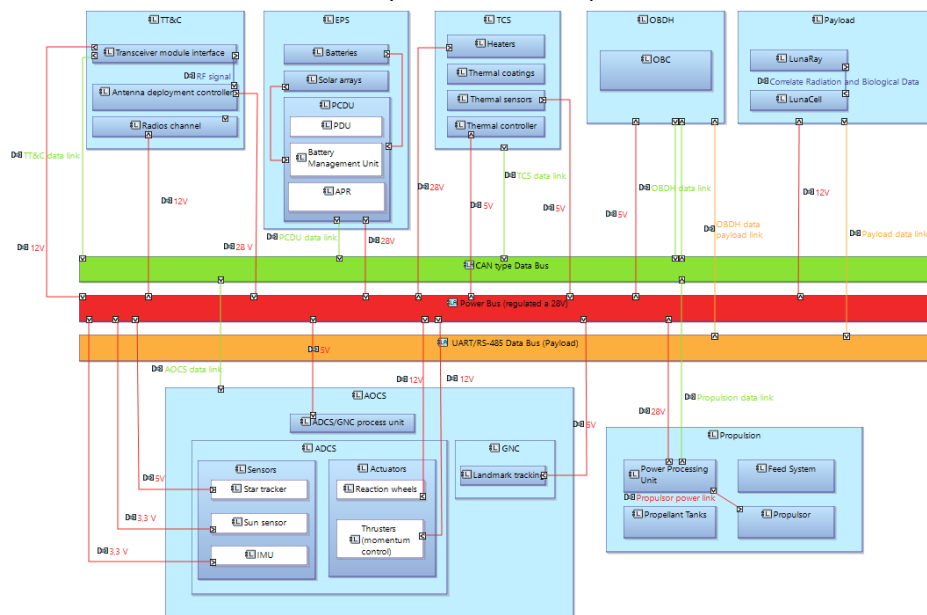


Figure 10-2: System Architecture Block Diagram

The EPS generates power through the solar arrays, conditions it in the PCDU, and distributes it across the spacecraft via the regulated Power Bus, while simultaneously charging the onboard batteries. The OBC, integrated within the OBDH subsystem, manages all command, telemetry, and payload data, using both the main Data Bus and a dedicated Payload Data Bus to prevent congestion and ensure priority handling of scientific measurements.

To support reliable communication, the spacecraft employs a dual-redundant CAN Bus interconnecting the primary subsystems, providing robustness against electrical noise and radiation-induced faults. Payloads interface through simpler UART links, which are sufficient for their lower bandwidth requirements. This hybrid architecture ensures consistent, faulty tolerant data exchange even under harsh space environmental conditions.

Figure 10-3 illustrates the hierarchical architecture of the spacecraft, providing an integrated overview of all major subsystems and their internal components, as well as the relationships that structure the overall system design.

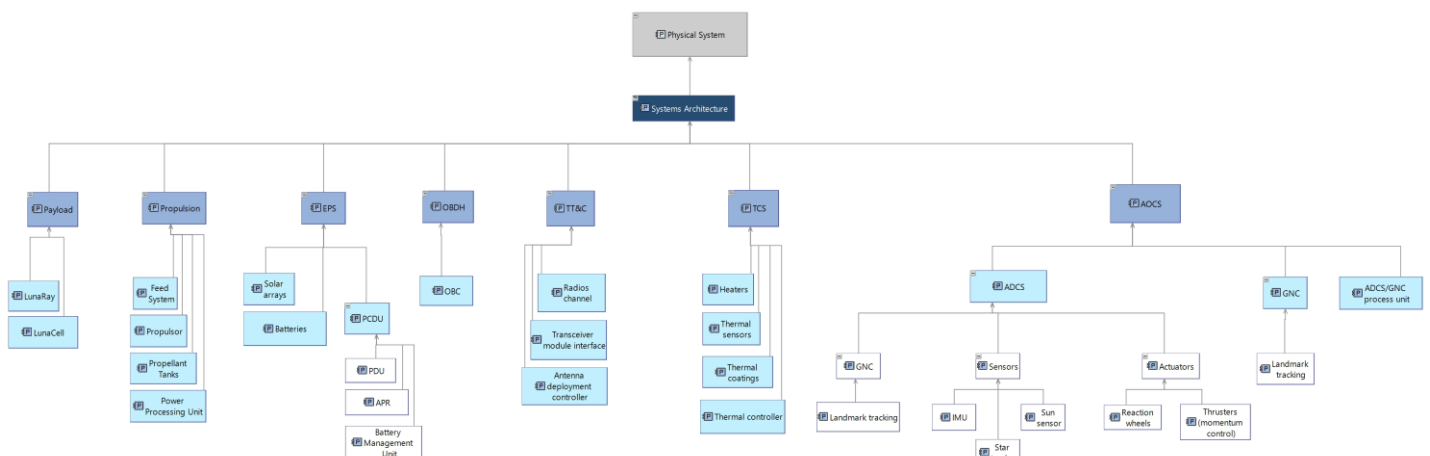


Figure 10-3: Product tree

10.4 Subsystem

10.4.1 Payloads

Table 10-2: Payloads Requirements

Requirement ID	Requirement	Type	Verification Method
R-SCI-07-00	LunaCELL shall allow repeated automatic measurements of absorbance of all liquid cultures with and without organisms, at 490 or 600 +/- 10 nm, at least 2 x per 24 h.	FUNC	Review, Analysis, Test
R-SCI-09-00	LunaRAY radiation data shall be made available also for LunaCELL data interpretation.	INTER	Review, Analysis
R-SCI-10-00	The biological payload shall not be exposed to a cumulative absorbed dose exceeding 300 Gy during the nominal operational phase of the mission.	FUNC	Review, Analysis
R-SCI-14-00	Following insertion into the operational orbit, a payload commissioning phase shall be completed and nominal scientific operations shall commence within 14 days.	FUNC	Review, Analysis

To fulfil the mission's SO, two complementary payloads are needed: LunaRAY and LunaCELL.

LunaRAY is devoted to the characterization of the lunar radiation environment and provides the dosimetry information required for the biological experiment. The instrument is based on two silicon detectors arranged in a telescope configuration, enabling measurement of particle count rates and absorbed dose rates, as well as reconstruction of the LET spectrum from coincidence events. By measuring the contribution of GCR, SPE and secondary particles generated by interactions with the lunar surface, LunaRAY enables the determination of absorbed dose and dose equivalent.

LunaCELL investigates the response of selected microorganisms exposed to the deep-space radiation environment. microorganisms are launched in a dormant state and activated in orbit by the injection of growth medium. Their proliferation is then monitored through periodic optical density measurements, making it possible to evaluate growth rates and survival capability under cumulative radiation exposure. By comparing shielded and unshielded samples and correlating the biological observations with the data acquired by LunaRAY, dose-response relationships can be established for the different species. The resulting dataset is expected to improve the understanding of radiation effects on biological systems and to support the development of bioregenerative life support systems for future long-duration exploration missions.

In accordance with the MDs and system requirements, four representative species were selected, covering the three domains of life and including a photosynthetically active organism relevant for BLSS.

- **Bacteria: *Bacillus subtilis*.** This bacterium synthesizes menaquinone-7 (vitamin K2), which promotes osteocalcin activation and plays an important role in bone metabolism. Furthermore, its ability to form endospores provides exceptional resistance to desiccation and extreme environmental conditions, making it well suited for space applications.
- **Archaea: *Methanosarcina barkeri*.** This archaeon produces methane by reducing carbon dioxide with hydrogen. Such metabolic capabilities make it particularly interesting for waste recycling and in-situ resource utilization, with potential applications in BLSS architectures.
- **Eukaryote: *Schizosaccharomyces pombe*.** This yeast exhibits DNA repair mechanisms closely resembling those of mammalian cells, making it a suitable model organism for investigating the effects of radiation on eukaryotic systems. Compared with the other selected species, it is expected to be the most sensitive to the lunar environment.
- **Photosynthetic organism: *Anabaena variabilis*.** This cyanobacterium contributes to oxygen production through photosynthesis and plays a key role in nitrogen fixation and recycling within closed-loop ecosystems. These characteristics make it a promising candidate for future BLSS applications.

Since the selected species exhibit different growth rates and life cycles, the optical density measurement frequency was increased from two to six measurements per day. This higher sampling rate allows for a more accurate reconstruction of the growth curves and a better characterization of the biological response to cumulative radiation exposure.

10.4.2 On Board Computer & Data Handling

The On-Board Computer (OBC) represents the central processing unit of the spacecraft and is responsible for command execution, data handling, subsystem monitoring, payload management, and communication with the orbiter. It coordinates all spacecraft operations, collects housekeeping telemetry, stores mission data, and distributes commands to the various subsystems.

At the current conceptual design stage, no specific flight-qualified computer has been selected. Therefore, OBC mass and power consumption estimates have been derived from representative CubeSat-class onboard computers with comparable processing capabilities and mission requirements. Candidate solutions considered during the preliminary trade-off include the GomSpace NanoMind Z7000, the EnduroSat OBC, and systems based on the Microchip PolarFire SoC architecture.

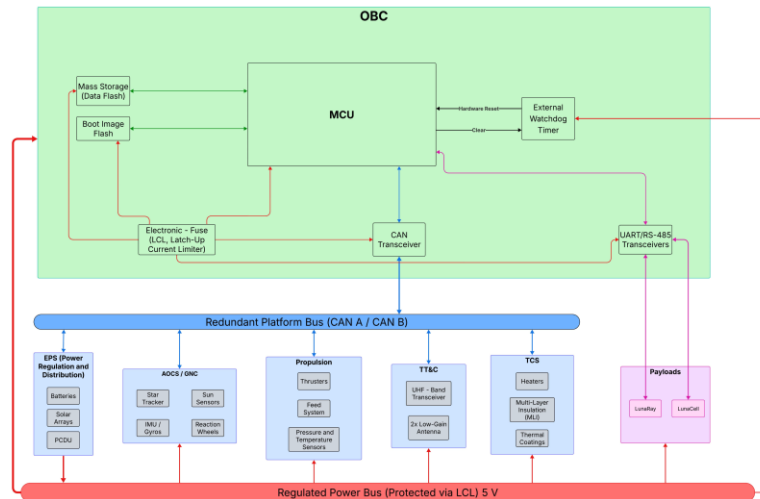


Figure 10-4: Proposed OBC Architecture and Interfaces with the Spacecraft Subsystems

Table 10-3: OBC Requirements

Requirement ID	Requirement	Type	Verification Method
R-OBDR-01-00	The OBDR subsystem shall accumulate 60 MB/week with sufficient margins for wear-leveiling and radiation-induced bad-block management.	FUNC	Review, Analysis
R-OBDR-01-01	The OBDR subsystem shall provide a non-volatile mass storage (Data Flash) capacity of at least 128 MB.	FUNC	Review, Analysis
R-OBDR-02-00	The OBDR subsystem shall include 32 MB of rad-tolerant Boot Flash to host three redundant On-Board Software (OBSW) images: Golden Image, Current Image, and Update Slot.	CONF	Review, Analysis
R-OBDR-03-00	The OBDR subsystem shall provide at least 4 MB of volatile working memory (SRAM) with active hardware EDAC protection against SEUs.	CONF	Review, Analysis
R-OBDR-04-00	The total power consumption of the OBDR subsystem shall not exceed 4 W during any nominal state.	FUNC	Review, Test

Since the mission operates in lunar orbit, the OBC must withstand a significantly harsher radiation environment. The architecture has therefore been designed according to two main principles: Fault Tolerance; Communication and Data-Flow Isolation.

At the core of the OBC is a radiation-tolerant MCU, responsible for spacecraft control and data processing. To mitigate radiation-induced failures, the OBC incorporates an electronic fuse with LCL functionality. This protection circuit continuously monitors the current absorbed by critical components and disconnects the affected branch in the event of a SEL, preventing permanent damage caused by excessive current consumption.

The memory subsystem is divided into two physically separated sections:

- The Boot Image Flash contains a protected golden image of the flight software that remains unchanged throughout the mission. This memory allows the spacecraft to recover from severe software corruption or radiation-induced upsets by restoring a verified operational configuration.
- The Mass Storage Flash is dedicated to mission data, payload outputs, telemetry logs, and operational software updates.

An independent hardware watchdog timer provides an additional layer of fault protection. Unlike software-based watchdogs, the external watchdog remains operational even if the MCU becomes unresponsive due to radiation-induced Single Event Upsets (SEUs) or software faults.

Communication between the OBC and the platform subsystems is performed through a redundant Controller Area Network (CAN) bus architecture composed of two independent channels (CAN A and CAN B). The CAN network connects EPS, ADCS, Propulsion Subsystem, TCS and TT&C. The selection of CAN is motivated by its high reliability, built-in error detection mechanisms (which increase communication robustness), deterministic behaviour, and widespread heritage in space applications. Differential signalling provides strong immunity against electromagnetic disturbances and radiation-induced transient effects.

The payloads are intentionally isolated from the main spacecraft CAN network and communicate directly with the OBC through dedicated UART interfaces implemented over an RS-485 physical layer. This architecture prevents high-volume payload data from saturating the spacecraft platform bus while maintaining reliable communication links. RS-485 was selected because its differential signalling significantly improves noise immunity compared with conventional single-ended serial interfaces.

10.4.3 Attitude and determination control system - ADCS

The ADCS architecture is heavily driven by the lunar operational environment. The negligible geomagnetic field precludes the use of magnetorquers, dictating a thruster-based momentum desaturation approach. The main objectives are maintaining continuous nadir pointing because of the observations reported in Section 9.2 (uninterrupted measurement of lunar albedo radiation by the LunaRAY payload) and executing off nadir slew manoeuvres for SK. The ADCS architecture was primarily driven by the following requirements, derived from the momentum budget analysis, the pointing accuracy assessment, and the attitude measurement accuracy analysis:

Table 10-4: ADCS Requirements

Requirement ID	Requirement	Type	Verification Method
R-ADCS-03-01	During landmark imaging, the ADCS attitude knowledge error shall be ≤ 14 arcsec (3σ).	FUNC	Review, Analysis
R-ADCS-03-02	During landmark imaging, the ADCS shall maintain relative pointing stability better than 14 arcsec per axis over any sliding time window of 0.061 s.	FUNC	Review, Analysis
R-ADCS-04-00	The ADCS shall provide a momentum storage capacity ≥ 0.04 Nms per axis.	FUNC	Review, Analysis
R-ADCS-06-00	During Delta-V manoeuvres, the ADCS shall align the spacecraft thrust axis (+X Body axis) with the target thrust vector commanded by the GNC subsystem, with an Absolute Pointing Error (APE) of ≤ 1 deg (3σ).	FUNC	Review, Analysis

To satisfy the pointing and stability requirements, the spacecraft employs a 3-axis stabilized, closed-loop control architecture. Attitude determination is governed by an Extended Kalman Filter (EKF). The EKF continuously propagates the spacecraft's attitude using high-frequency angular rates from the IMUs, correcting the inherent inertial drift via periodic high-precision quaternion updates from the Star Trackers. During Safe/Survival modes or initial detumbling, when Star Trackers are offline, the EKF falls back on Sun Sensor data for coarse Sun-pointing.

Two types of Sun Sensors are implemented: an array of 6 Coarse Sun Sensors (CSS) providing attitude knowledge up to 15 deg (3σ), and two Fine Sun Sensors (FSS) with an accuracy of up to 0.3 deg (3σ). The CSS are used for detumbling and act as a redundant backup to the FSS, which serve as the primary sun sensors. The Star Trackers provide the primary attitude determination data during all other mission phases. To ensure continuous sky coverage and prevent simultaneous blinding from the Sun or lunar albedo, the three units are distributed across the spacecraft faces, with one directly aligned along the -Z body axis (Zenith). This configuration yields a combined attitude knowledge accuracy of 6.4 arcsec (3σ) (~ 0.0018 deg) in all directions. Regarding the IMUs, one unit is continuously active within the control loop while a second is kept in cold redundancy. Their function is to continuously propagate attitude knowledge, dampen angular rates during manoeuvres, and maintain the required pointing stability during the LTC imaging process.

The complete performances for the attitude sensors are reported in Table 10-5.

Table 10-5: Attitude Sensors Performances

Star tracker	CSS	FSS	IMU
Arcsec: TWINKLE	Tensor Tech: CSS-10	Tensor Tesch: Tensor FSS-30D	Safran: STIM318
Cross-Boresight: 3" (1σ) About-Boresight: 30" (1σ)	FOV: 120 deg Acc: 5 deg (1σ)	FOV: 120 deg Acc: 0.1 deg (1σ)	Dynamic Range: ± 400 deg/s Bias Instab. 0.3 deg/h Angular Rand. Walk: 0.15 deg/Vh

Regarding the actuation system, primary attitude control is provided by four BCT RWp050 reaction wheels arranged in a tetrahedral configuration. With a maximum storage capacity of 50 mNms and a peak torque of 7 mNm, the wheels

are sized to absorb accumulated environmental torques over approximately 24-hour desaturation cycles across the release, operational, and graveyard orbits. To perform the initial detumbling and manage periodic reaction wheel desaturation, an 8-thruster RCS is employed. These ECAPS micro-thrusters deliver 100 mN of thrust and utilize LMP-103S green monopropellant, specifically selected for its high specific impulse ($I_{sp} > 190$ s) to significantly minimize the subsystem's wet mass compared to cold-gas alternatives.

The block diagram in Figure 10-5 illustrates the general ADCS architecture. The Guidance module uses GNC navigation data (orbital position, velocity, and time) to compute the reference attitude (needed for continuous nadir pointing) as a desired quaternion and angular rate vector. The resulting attitude error drives the Attitude Controller, which computes a commanded torque executed by the Actuator Plant (reaction wheels and thrusters). This control torque is algebraically summed with external disturbance torques to yield the effective torque governing the spacecraft's true rotational dynamics.

To close the loop, Attitude Determination Sensors measure the physical state and, since measurements are corrupted by noise and bias, pass the raw data to a State Estimator (EKF), which returns a clean estimated state to the controller. Concurrently, the Actuator Plant feeds reaction wheel angular velocity and accumulated momentum to both the controller, for momentum monitoring and desaturation scheduling, and the EKF, which relies on this data to correctly propagate the spacecraft's dynamics. Without accurate wheel state information, the estimator would misinterpret internal torque exchanges as unknown external disturbances, yielding inaccurate attitude estimates.

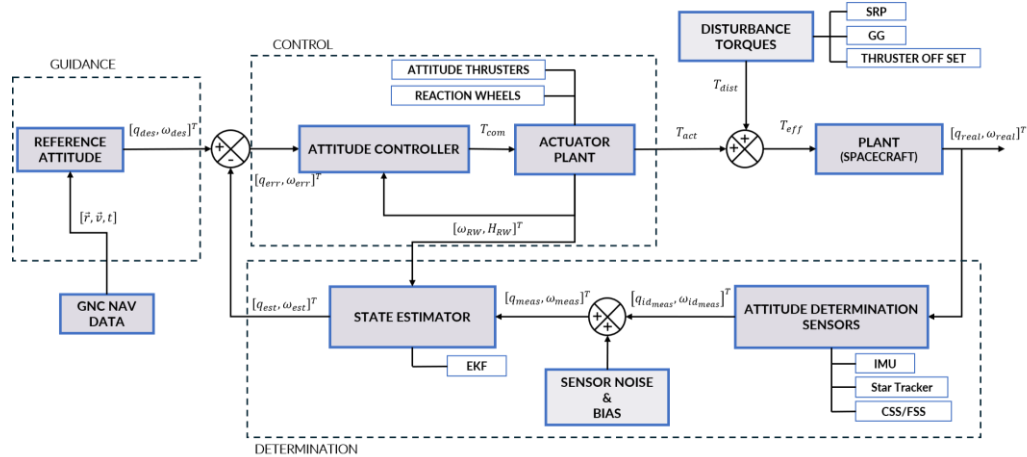


Figure 10-5: ADCS Block Diagram

10.4.4 GNC & Propulsion System

The GNC and Propulsion subsystem have the fundamental role of ensuring mission success through precise orbit control, execution of nominal manoeuvres, and autonomous fault management. The subsystem architecture was defined starting from payload-driven functional requirements and high-level system constraints. The primary driver for sizing the propulsion system is the total mission ΔV budget, 909.84 m/s where an additional 10% margin is applied (R-PRO-04-00). This budget is divided into three main phases: orbital transfer (322 m/s), station keeping (109 m/s) and final disposal (478 m/s). This requirement, combined with the stringent mass and volume constraints has led to the selection of an electric propulsion system. For the GNC subsystem, the main driver is the need to ensure accurate position estimation in the lunar environment, which is characterized by the absence of GNSS signals.

Table 10-6: GNC & Propulsion Requirements

Requirement ID	Requirement	Type	Verification method
R-PRO-04-00	The propulsion system shall provide a minimum total mission delta V of 993.94 m/s to support lunar orbit insertion, station keeping and disposal.	OPER	Review, Analysis
R-PRO-06-00	The main thruster shall maintain a Specific Impulse better or equal to 1000 s during all nominal orbital transfer manoeuvres.	FUNC	Review, Analysis
R-PRO-08-00	The propulsion system shall not exceed a maximum power draw of 160 W in High-Performance mode.	FUNC	Review, Analysis
R-PRO-10-00	The main thruster shall be qualified for an operational life of at least 10,000 hours to ensure mission success in the lunar gravity field.	FUNC	Review, Analysis

R-GNC-02-01	The system shall determine the satellite's position with a relative navigation error not exceeding 600 m along the orbit.	FUNC	Review, Analysis
R-GNC-03-00	In the eventuality of loss of visual references, the system shall maintain the position estimation via inertial propagation with a maximum drift not exceeding 556 m per 26 minutes of autonomous navigation.	FUNC	Review, Analysis
R-GNC-05-00	During orbital transfer manoeuvres, the control system shall deliver the ΔV planned by the Lambert solver with a maximum error of 0.5% in magnitude and 0.5° in direction.	FUNC	Review, Analysis

To ensure continuous state estimation and maximize fault tolerance, the subsystem has been designed around three primary navigation strategies, which are dynamically selected according to the mission phase and prevailing operational conditions.

The control architecture operates in open-loop, propagating the state without closed-loop feedback correction.

The hardware supporting the implementation of these three navigation strategies consists of two primary sensors: an IMU, shared with the ADCS subsystem, and a landmark-tracking camera. A LTC IM200 with a 50 mm lens has been selected to acquire surface imagery. The camera provides a 12.9° FOV and achieves a cross-boresight accuracy of 13.3 arcsec, enabling precise landmark-based navigation and attitude determination.

Three navigation strategies are dynamically selected based on mission phase and operational conditions. The **Optical Navigation strategy** (primary) relies on the LTC to estimate the spacecraft position through surface feature observations, with the IMU and orbital propagator providing continuous state propagation between measurements; this strategy achieves a navigation accuracy of 70–600 m (according to altitude). During lunar eclipses, when optical observations are unavailable, **Dead Reckoning** takes over: position and velocity are propagated using the perturbed equations of motion and IMU measurements only, with navigation uncertainty increasing gradually over time, a drift of 560 m over a 25-minute eclipse period therefore, the shorter the observation gap, the lower the degradation in accuracy. Finally, **Autonomous Safe Navigation** is activated in SAFE and SURVIVAL modes, where all sensors are switched off and only the orbital propagator remains active, integrating the equations of motion from the last available state vector; no manoeuvres are permitted, leading to continuous accumulation of navigation uncertainty.

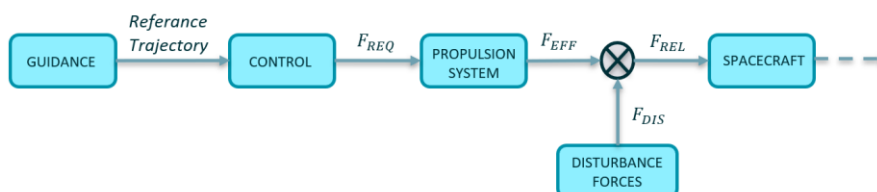


Figure 10-6: GNC Architecture

Due to the high delta-V demanded by the mission, chemical propulsion systems were ruled out during the preliminary design phase owing to their inherently low specific impulse. A trade-off analysis was therefore conducted focusing on electric propulsion technologies available for CubeSat evaluated against five normalized figures of merit: specific impulse, specific mass, power-to-thrust ratio, thrust level, and overall efficiency. The Hall Effect thruster emerged as the optimal solution, offering the best balance between high specific impulse, a thrust density superior to ion engines, and a power draw compatible with the platform power budget. Furthermore, its compact dimensions (8 x 5.5 cm) and the low mass (1.16 kg) make it fully compatible with a 12U CubeSat. Xenon was selected as the propellant working fluid by virtue of its high atomic mass, low ionization energy, chemical inertness, and complete non-toxicity, the latter being a important requirement given the presence of biological payload on board. The chosen commercial thruster is the Busek BHT-100, a miniaturized unit featuring advanced magnetic shielding, qualified for CubeSat applications and derived from the flight-heritage BHT-200 heritage model. The technical specifications are summarized in Table. The total propellant mass required by the mission is 1.709 kg. The required propellant mass was calculated using the rocket equation, based on the satellite's dry mass and the total required delta-V. Applying a 1.18 margin factor, it was decided to load 2 kg of propellant, pressurized in a tank at 150 bar.

Table 10-7: Engine and Feed System Specifications

BUSEK BHT-100	
Specific Impulse	1100 s
Thrust	7 mN
Discharge power	100 W
Thruster weight	1.16 kg
Efficiency	0.4
Propellant	Xenon

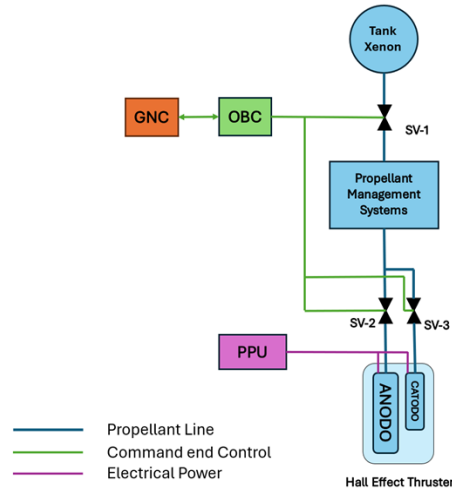


Figure 10-7: Propulsion System Architecture

10.4.5 Electrical Power System – EPS

The EPS is one of the critical subsystems of the CubeSat, since it is responsible for generating, storing, conditioning, distributing and protecting the electrical power required by all onboard units throughout the mission. The primary requirements are:

Table 10-8: EPS Requirements

Requirement ID	Requirement	Type	Verification method
R-EPS-04-01	The heater distribution lines shall be protected by dedicated HLCLs.	DES	Review, Analysis
R-EPS-09-00	The solar array sizing shall account for radiation-induced degradation of the solar cells throughout the mission lifetime.	DES	Review, Analysis
R-EPS-10-00	The solar cells shall have an efficiency of at least 29,5% at BOL.	PHYS	Review, Analysis
R-EPS-11-00	The batteries shall have at maximum 40% of Depth of Discharge (DoD).	DES	Review, Analysis

The first two requirements were selected to be consistent with ECSS standards. The third requirement reported here was derived from a state-of-the-art study of CubeSat missions designed for lunar orbit/deep-space missions. To be consistent with these missions, it was decided to impose as a requirement a solar cell efficiency at BOL of at least 29.5%. This requirement led to the identification of the most suitable solar cell technology for this type of mission, namely triple-junction GaAs cells. Regarding the fourth requirement, a maximum DoD value of 40% was selected for the battery sizing. This value is consistent with missions of a duration similar to ours and, in addition, a lower value would have led to an excessive oversizing of the battery pack. The sizing of the solar panels and batteries was performed by first calculating the Power Budget. As a result, the power consumption of each operational mode, both in eclipse and in sunlight, was obtained and is shown in Figure 10-8.

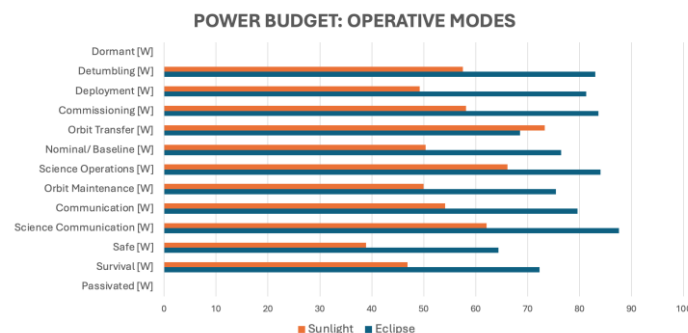


Figure 10-8: Power Budget

The peak power required in eclipse and in sunlight is 79,236 W and 73,270 W, respectively. These values were used as inputs for the sizing of the solar panels and batteries. For the panels, a total power required during daylight to power the spacecraft over the entire orbit of 158,05 W was obtained, achievable with a total panel area of 0.525 m². For the

batteries, instead, a total required capacity of 236,3 Wh was obtained. The values obtained from the sizing consider several factors; the main ones for the panels are:

- Ideal solar cell efficiency: 30%
- Solar cell degradation per year: 3%
- Inherent degradation: 77%

Whereas for the batteries:

- Depth of Discharge: 40%
- Battery efficiency: 95%

Regarding the Energy Budget, it was decided to analyse the battery charge and discharge cycle during our worst case. It was selected one day during which the spacecraft is in Science Operation Mode (the OM with the maximum average power) and spends the longest time in eclipse, a 68-minute eclipse. In this analysis, a sunlight sun angle of 0° was always considered, as sun-tracking panels enabled by a SADA device were selected. This choice was made due to the high-power demand of the thruster. The graph shows the trend of the SoC during the various orbits performed on this particularly critical day. From the graph, it can be observed that the SoC never falls below 67%; therefore, the DoD percentage always remains below 40%, as required.

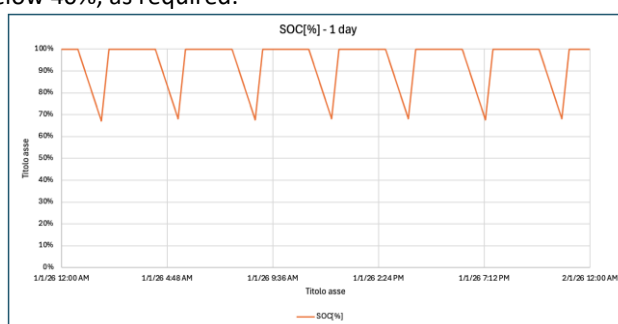


Figure 10-9: SoC% during 1 day in Science Operation Mode (worst case)

The architecture selected for the EPS is a distributed architecture and, regarding the solar panel interface, the selected regulation strategy is MPPT, or maximum power point tracking. With this regulation method, it is possible to maximize the efficiency of the solar panels by always operating at the maximum power point of the panel, which is a fundamental choice since the spacecraft operates in deep space and is subject to extreme thermal excursions. For the main power bus, a regulated 28 V bus was selected. This choice was made in view of the use of the BHT-100 thruster, which in nominal operation requires a continuous power consumption of 100 W. The regulated 28 V bus guarantees a stable and predictable input to the thruster PPU. Regarding the batteries, it was decided not to use a single 300 Wh battery pack, but rather three 100 Wh battery packs connected in parallel, each with its own battery management unit, to comply with ECSS standards. In this way, single points of failure are avoided. The PDU manages the power distribution from the main power bus to the various subsystems through dedicated 3.3 V, 5 V, 12 V and 28 V channels. The voltages were selected according to the requirements of the individual subsystems. Each power rail has a DC/DC converter, excluding the 28 V rails, and a dedicated LCL. To comply with ECSS standards, the heater line uses an HLCL rather than a simple LCL. In conclusion, the main components of our EPS are summarized in **Error! Reference source not found.0-9**.

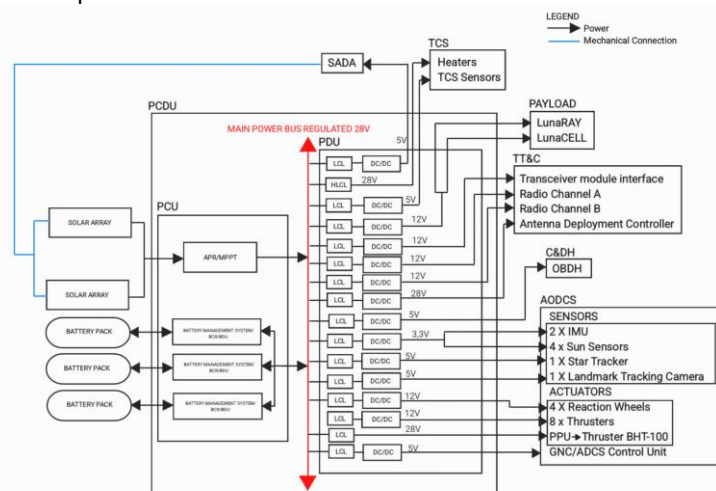


Figure 10-10: EPS Architecture

Table 10-9: EPS components

COMPONENTS	DESCRIPTION	REFERENCES
SOLAR ARRAY	2 deployable wings, sun-tracking, 12 solar panel in total Power produced at BOL: 320,07 W Power produced at EOL: 301,15W	MMA Design-eHaWK Azur Space 3G30C
BATTERY	3 battery pack in parallel (8S1P) Battery capacity: 300Wh; 10,5Ah	GomSpace NanoPower BP8 100Wh
POWER MANAGEMENT	1 PCU board (1APR and 3 battery management unit) and 1 PDU board	NanoAvionics EPS 2.0

10.4.6 Communication System

The primary objective of the Communication System is to establish and maintain a robust and highly reliable communication link between the CubeSats and the Mothercraft. Given the harsh lunar radiation and thermal environment, ensuring link availability is critical to preventing mission failure and maintaining continuous operational capabilities for biological and environmental characterization. By utilising the Orbiter as a dedicated radio relay network architecture, the subsystem minimizes on-board hardware complexity and RF power consumption while guaranteeing that scientific payloads and housekeeping data are safely downloaded. The system drivers focus on maximizing signal integrity, architectural simplicity, and omnidirectional fault tolerance to ensure link closure even during non-nominal attitudes or emergency safe modes. The table below outlines the core requirements for the preliminary design phase.

Table 10-10: Communication Requirements

Requirement ID	Requirement	Type	Verification method
R-COMM-01-00	The system shall operate in the VHF band at 145 MHz for the Uplink channel to receive commands from the Orbiter.	OPER	Review, Analysis
R-COMM-02-00	The system shall operate in the UHF band at 437 MHz for the Downlink channel to transfer payload and telemetry data.	OPER	Review, Analysis
R-COMM-03-00	The system shall complete the daily data transfer within the minimum access time of 9350 seconds.	OPER	Review, Analysis
R-COMM-04-00	The system shall support BPSK modulation for every communication link.	FUNC	Review, Analysis
R-COMM-05-00	The communication subsystem shall maintain a minimum link margin of 3 dB under nominal operating conditions.	FUNC	Review, Analysis
R-COMM-06-00	The transceiver shall support selectable data rates of 1200 bps for uplink and 9600 bps for nominal Downlink.	FUNC	Review, Analysis

The communication architecture relies on a dual cold/operational redundancy configuration featuring two completely independent radio chains feeding two distinct antenna systems to eliminate single points of failure. The hardware entirely consists of space-proven Commercial Off-The-Shelf (COTS) components: the ISISpace VHF/UHF Transceiver integrated with the ISISpace Deployable Antenna System. To ensure spherical coverage and eliminate radiation pattern nulls caused by physical chassis shielding, the two antenna modules are installed on opposite faces of the CubeSat structure. For the Uplink, a VHF monopole configuration operating at a nominal frequency of 145 MHz is utilized, whereas the Downlink employs a four-element UHF turnstile configuration operating at 437 MHz. To ensure robust signal integrity during the transmission of the consolidated daily data volume of 4.85 MB, the system implements BPSK modulation and encapsulates data packets according to the standard AX.25 protocol. The transceiver supports selectable transmission data rates, utilizing 9600 bps for nominal downlink data dumps and scaling down to 1200 bps for uplink commands and emergency mode. During operations, the OBC calculates the optimal geometric tracking vector toward the Orbiter and routes the signal to activate the antenna module that has the best visibility window. Passive VHF/UHF diplexers are integrated into each chain to separate the weak received uplink signals from the high-power downlink transmission, enabling full-duplex operation and preventing receiver saturation. The radio modules are housed near the antenna arrays to limit insertion losses to 1.5dB, and electromagnetic shielding is implemented to protect the sensitive OBC electronics from RF interference. To manage resources, a strict operational duty cycle of 5% is enforced, resulting in a daily transmission time of 70 minutes and 30 seconds. The antenna deployment is triggered by redundant burn wires requiring 2W of electrical power during activation.

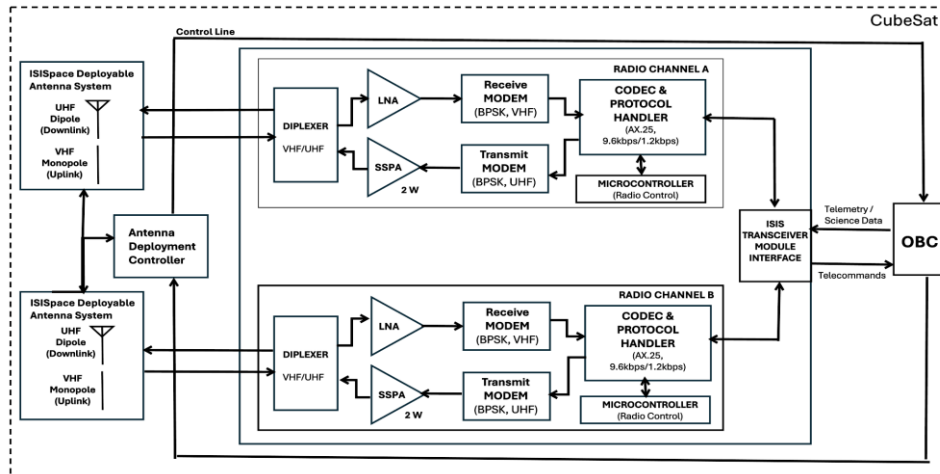


Figure 10-11: Communication System Architecture

10.4.6.1 Link and Data Budgets

To verify the technical feasibility of the selected configuration, link budgets were calculated for both the uplink and downlink paths based on a worst-case slant range of 8100 km. Both channels employ BPSK modulation and account for 1.5dB of line losses. The nominal downlink operates at 437 MHz with a data rate of 9600 bps, while the uplink operates at 145 MHz and 1200 bps. A worst-case 90° pointing error was considered for the uplink to simulate emergency conditions.

The resulting margins both exceed the minimum design requirement of 3 dB. Therefore, positive link closure is ensured under both nominal and non-nominal operating conditions.

Table 10-11: Link Budget

Uplink (Orbiter to CubeSat)		Uplink (Orbiter to CubeSat)	
EIRP	3.66 dB	EIRP	5.51 dB
Space loss	-163.43 dB	Space loss	-153.84 dB
G/T	-7.80 dB	G/T	-25.73 dB
Antenna pointing losses	-0.06 dB	Antenna pointing losses	-4.70 dB
Eb/No	21.15 dB	Eb/No	19.04 dB
C/No	60.97 dB	C/No	49.83 dB
Implementation loss	-2.00 dB	Implementation loss	-2.00 dB
Margin	8.65 dB	Margin	6.54 dB

The DB consists of two main contributions: housekeeping (HK) data and mission data. The estimation of the HK data volume was based on the identification of more than 250 telemetry parameters, each characterized by its corresponding bit size and sampling frequency. Since subsystem activity varies depending on the OM, a dedicated HK budget was established for each mode. This approach made it possible to identify the most demanding operational scenario in terms of data generation, which was subsequently used to size the onboard mass memory.

The detailed definition of the telemetry parameters also enabled an accurate estimation of the communication overhead associated with packet headers, timestamps, and protocol structures required by ESA standards. Finally, an additional margin was introduced to account for the uncertainties inherent to a Phase 0/A design.

Table 10-12: Data Budget

Data	Raw	Overhead (+30%)	Margin (+20%)	Total
Mission	192 kB/day	57.6 kB/day	49.94 kB/day	300 kB/day
Housekeeping	2.92 MB/day	0.876 MB/day	0.754 MB/day	4.55 MB/day
				4.85 MB/day

10.4.7 Thermal Control System – TCS

The primary objective of the Thermal Control System (TCS) is to maintain all spacecraft subsystems within their safe Operational and Survival Temperature limits throughout all mission phases.

For the preliminary mission analysis phase, a Single-Node lumped parameter thermal model was developed. The main spacecraft body was approximated as an isothermal sphere with an external radiative surface area of 0.36 m² (excluding stowed external deployable arrays).

The global allowable operational temperature range for the primary spacecraft body is bounded deterministically by the overlap of the most stringent subsystem limits, resulting in a required core thermal envelope of +20°C to +40°C.

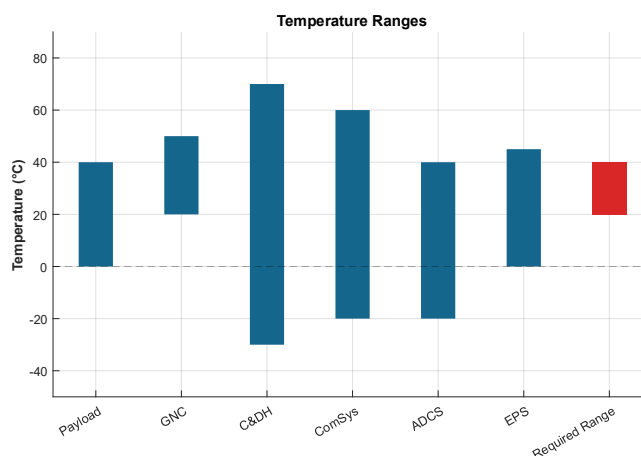


Figure 10-12: Temperature Ranges

The internal heat dissipation from electronic components serves as the primary internal thermal source, evaluated under two bounding operational conditions:

- Maximum Power Consumption (Hot Case): 73.3 W
- Minimum Power Consumption (Cold Case/Eclipse): 39.52 W

Utilizing a white paint coating to maximize passive radiative cooling into deep space during the Hot Case, while actively managing Cold Case drops via internal heaters.

The selected coating baseline is S13G-LO white paint, exhibiting the following stable thermo-optical properties:

- Solar Absorptivity (α): 0.20
- Thermal Emissivity (ϵ): 0.92

Under this configuration, the environmental external heat flux inputs to the primary body are:

- Absorbed Solar Power: 25.3 W
- Absorbed Lunar Albedo Flux: 1.2 W
- Maximum Absorbed Lunar Infrared (IR) Flux: 59.0 W

The environmental and internal dissipation inputs yield the following steady-state and transient equilibrium core temperatures:

- Maximum Equilibrium Temperature (Hot Case): +34.7°C
- Minimum Equilibrium Temperature (Cold Case - Passive): -42.6°C

To achieve this compliant cold-case posture during the worst-case 124-minute lunar eclipse, the active TCS requires a total internal heating power of 17 W. This will be supplied by localized patch heaters distributed over the most sensitive nodes.

Due to their external stowed configuration and low thermal mass, the solar panels are subjected to extreme temperature fluctuations:

- Maximum temperature reached: approximately +40°C
- Minimum temperature reached during eclipse periods: approximately -175°C

The implementation of dedicated solar panel heaters is deferred to a future design iteration, pending confirmation on whether the solar cells require a minimum operating temperature higher than the temperature reached during the worst-case cold condition. Additionally, future design loops will explore optimizing low-conductivity mechanical hinges between the primary body and the panel structures to minimize heat leakage during deep eclipse phases.

10.4.8 Structures

The structural subsystem is designed to maintain the mechanical integrity of the spacecraft, protect internal subsystems from the harsh cis-lunar environment, and guarantee a standard, interference-free interface with the deployment canister.

The configuration and functional constraints are dictated by the launch vehicle parameters and the EXOpod Nova 12UXL deployer specifications:

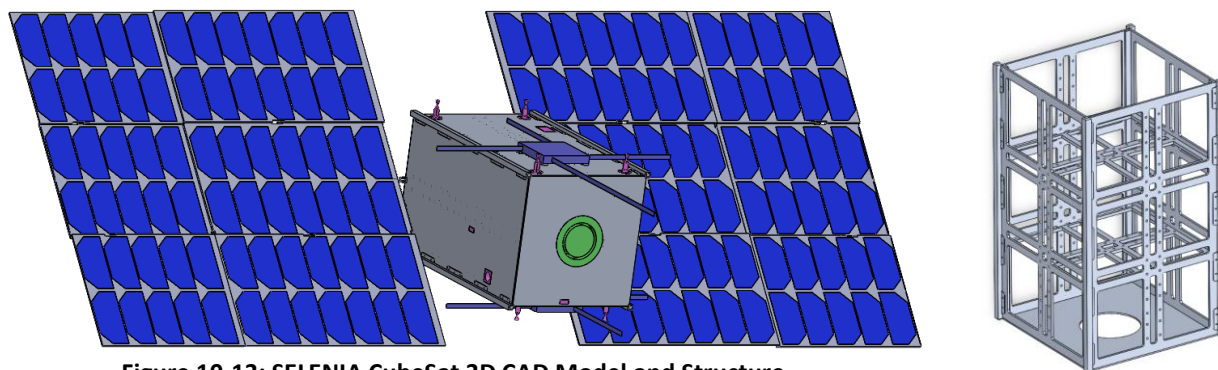
Table 10-13: Structures Requirements

Requirement ID	Requirement	Type	Verification method
R-STRU-01-00	The system's total mass shall be less than 100kg (for the 4-satellite constellation).	PHYS	Review, Analysis
R-STRU-02-00	The volume of each satellite shall not exceed 16U.	PHYS	Review, Analysis
R-STRU-03-00	The system's centre of mass shall lie within a ± 45 mm range along the X and Y axes, measured from the geometric centre.	PHYS	Review, Analysis
R-STRU-03-01	The system's centre of mass shall lie within a ± 70 mm range along the Z axis, measured from the geometric centre.	PHYS	Review, Analysis
R-STRU-06-00	All deployable elements shall be retained by the CubeSat structure.	DES	Review, Inspection
R-STRU-07-01	The rails shall provide a flat contact area greater than 6.5 mm $\times$ 6.5 mm.	PHYS	Review, Inspection
R-STRU-07-02	The rails shall be made of aluminium treated with Type III hard anodizing.	DES	Review
R-STRU-07-03	The minimum rail width shall be at least 8.5 mm.	PHYS	Review, Inspection
R-STRU-07-05	If present, external elements shall protrude no more than 39.5 mm beyond the rail plane.	PHYS	Review, Inspection
R-STRU-07-06	The external rail-to-rail distance along the X and Y axes shall be 226.3 mm $\pm$ 0.1 mm.	PHYS	Review, Inspection
R-STRU-07-07	the rail length along the Z axis shall be 365.9 mm $\pm$ 0.5 mm.	PHYS	Review, Inspection

The structural design prioritises modularity, manufacturing simplicity, and cost reduction. The architecture is subdivided into four independent longitudinal rails, treated with Type III Hard Anodization to prevent cold welding within the deployer canister without affecting the remaining chassis, four identical skeleton panels with standardised hole grids for internal mounting, and outer closing plates providing thermal and particulate contamination shielding. The entire primary structure is manufactured from Aluminium 6061-T6, selected for its flight heritage, high structural stiffness, and favourable strength-to-weight ratio to meet mass constraints.

Internal packaging and external appendage layouts have been optimised to balance the CoM, maximise volumetric efficiency, and minimise subsystem cross-interference. The propellant tank is integrated adjacent to the thruster core to shorten feed-system lines while preserving an internal volumetric corridor. The reaction wheel assembly is placed near the geometric centre to reduce disturbance torques during momentum dumping. Both scientific payloads are located at the opposite end of the propulsion module to isolate them from electromagnetic disturbances and plume contamination, with LunaRAY specifically mounted on the nadir-pointing face. Attitude control thrusters, patch antennas, and solar arrays are distributed on alternating faces to avoid operational envelope overlap; the solar arrays are positioned on the face opposite the main engine to limit plume and EMI interactions.

The resulting 3D CAD configuration meets the envelope constraints of the EXOpod Nova 12UXL deployer.


Figure 10-13: SELENIA CubeSat 3D CAD Model and Structure

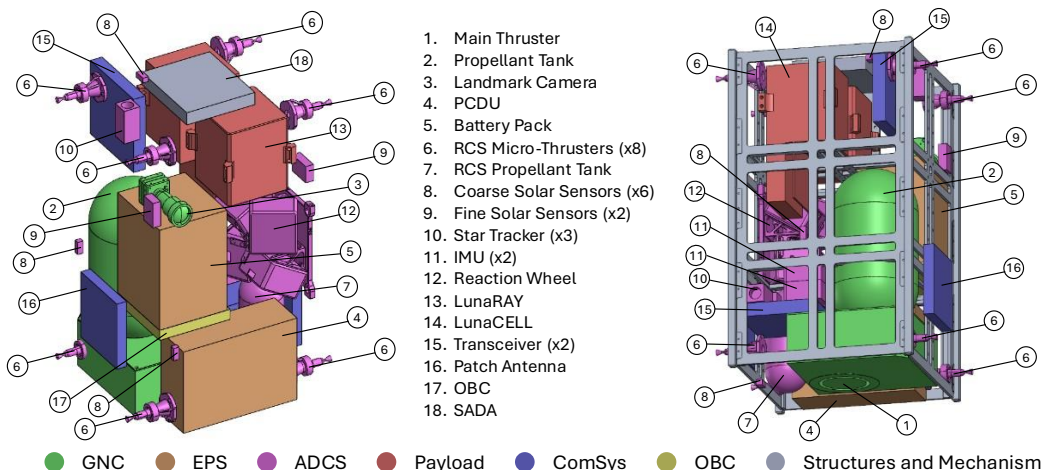


Figure 10-14: SELENIA CubeSat 3D CAD Model

10.4.9 Mass and Volume Budgets

The spacecraft mass budget, presented in Table -14, has been established in strict accordance with the System Margin Policy for ESA IOD CubeSat Projects. The final single spacecraft wet mass, including system margin, is 24.72 kg, for a total constellation mission mass of 98.88 kg across 4 units (with a margin of 20%).

Table 10-14: SELENIA CubeSat Mass Budget

Component / Subsystem	Nominal Mass (kg)	Margin (%)	Margin (kg)	Mass with Margin (kg)	% on Total (Wet)
LunaRAY (Payload)	0.30	5.00%	0.015	0.32	1.3%
LunaCELL (Payload)	0.60	5.00%	0.030	0.63	2.5%
Structure and Mechanism	3.20	10.00%	0.320	3.52	14.2%
EPS (Batteries & PCDU)	4.30	5.00%	0.215	4.52	18.3%
Propulsion (Dry mass)	4.00	5.00%	0.200	4.20	17.0%
ADCS	2.40	5.00%	0.120	2.52	10.2%
C&DH	1.50	10.00%	0.150	1.65	6.7%
TT&C	1.50	5.00%	0.075	1.58	6.4%
TCS	0.80	10.00%	0.080	0.88	3.6%
TOTAL (Dry Mass)	18.60	20.00%	3.720	22.32	90.3%
Propellant	2.00	5.00%	0.100	2.10	8.5%
TOTAL (WET)	20.60	20.00%	4.120	24.72	100.0%

The maximum reference volume for all subsystem packaging is defined by the satellite's internal structural volume of 15.04 L. Beyond the standard component-level design margins, a 15% integration contingency has been added to the baseline CAD volumes. This allows the budget to realistically account for internal routing elements, including wiring harnesses, connectors and fluid lines, before computing the final percentage impact of each subsystem.

Table 10-15: SELENIA CubeSat Volume Budget

Component	Volume (L)	Volume with Cables (L)	Margin (%)	Margin (L)	Volume with margin (L)	% on total (15.04 L)
LunaRAY	0,54	0,62	5,00%	0,031	0,65	4,3%
LunaCELL	0,64	0,74	5,00%	0,037	0,77	5,1%
Structure and Mechanism	1,06	1,22	10,00%	0,122	1,34	8,9%
EPS	1,71	1,97	5,00%	0,098	2,06	13,7%
Propulsion	2,28	2,62	5,00%	0,131	2,75	18,3%
ADCS	0,60	0,69	5,00%	0,035	0,72	4,8%
C&DH	0,10	0,12	10,00%	0,012	0,13	0,8%
TT&C	0,42	0,48	5,00%	0,024	0,51	3,4%
TCS	0,30	0,35	10,00%	0,035	0,38	2,5%
TOTAL	7,65	8,80	20,00%	1,760	10,56	70,2%

11 System Development

11.1 Assembly, Integration and Verification Plan

11.1.1 Verification Methods end level

The verification approach was defined in accordance with the ECSS standards, as specified in [AD07]. It consists of four methods used to verify the requirements: Inspection, Review of Design, Analysis, and Test. The verification method was identified for each requirement in the corresponding subsystem section. Table 11-1 summarizes the requirements related to the AIV Plan.

Table 11-1: AIV Requirements

Requirement ID	Requirement	Type	Verification Method
R-SYS-08-00	The spacecraft electrical interface with the launch vehicle shall be verified throughout the launch campaign to ensure correct spacecraft-launcher integration.	INTER	Analysis, Test
R-SYS-09-00	The spacecraft RF configuration shall remain unchanged from H0–1h30 until 20 seconds after spacecraft separation.	FUNC	Review
R-SYS-10-00	All spacecraft materials and surface finishes shall comply with the outgassing requirements specified in the Ariane 6 User's Manual, including the applicable limits for TML and CVM.	DES	Review, Analysis
R-SYS-11-00	Recurring FM of the constellation shall be subjected to acceptance testing, while qualification shall be demonstrated by heritage from the PFM.	DES	Review, Analysis
R-SYS-12-00	The spacecraft structure shall withstand the quasi-static launch loads specified by the Ariane 6 launch vehicle without experiencing structural failure or degradation of functionality.	ENV	Analysis, Test

11.1.2 Model philosophy

The most suitable model philosophy—Prototype, ProtoFlight, or Hybrid—was selected through a trade-off analysis based on five FoMs:

- Cost:** minimize and optimize the number of models while keeping the overall mission cost under control.
- Schedule:** meet the launch constraint defined in [R-MIS-02-00] by optimizing the design, manufacturing, and verification timelines.
- Verification robustness:** provide an adequate level of confidence in system compliance and performance.
- Risk reduction:** identify critical issues at an early stage, preventing them from emerging during later development phases, when their impact on cost and schedule would be more significant.
- Implementation effort:** limit the complexity involved in applying the selected verification approach.

Model Philosophy Trade-off Matrix

Criterion	Weight	Rationale	Scoring Scale		Meaning
Cost	20%	Higher score = lower cost / more affordable	5	Excellent / very favourable	
Schedule	20%	Higher score = shorter and more manageable A/V timeline	4	Good / favourable	
Verification Robustness	25%	Higher score = stronger verification coverage and confidence	3	Acceptable / medium	
Risk Reduction	25%	Higher score = lower probability of late design/interface issues	2	Weak / unfavourable	
Implementation Effort	10%	Higher score = easier implementation and lower operational complexity	1	Poor / not favourable	
TOTAL	100%	Must equal 100%			

Model Philosophy	Cost	Schedule	Verification Robustness	Risk Reduction	Implementation Effort	Weighted Score
Prototype	1	1	5	5	1	3.00
Protoflight	5	5	2	1	5	3.25
Hybrid Approach	4	4	4	4	3	3.75

Figure 11-1: Model Philosophy trade-off

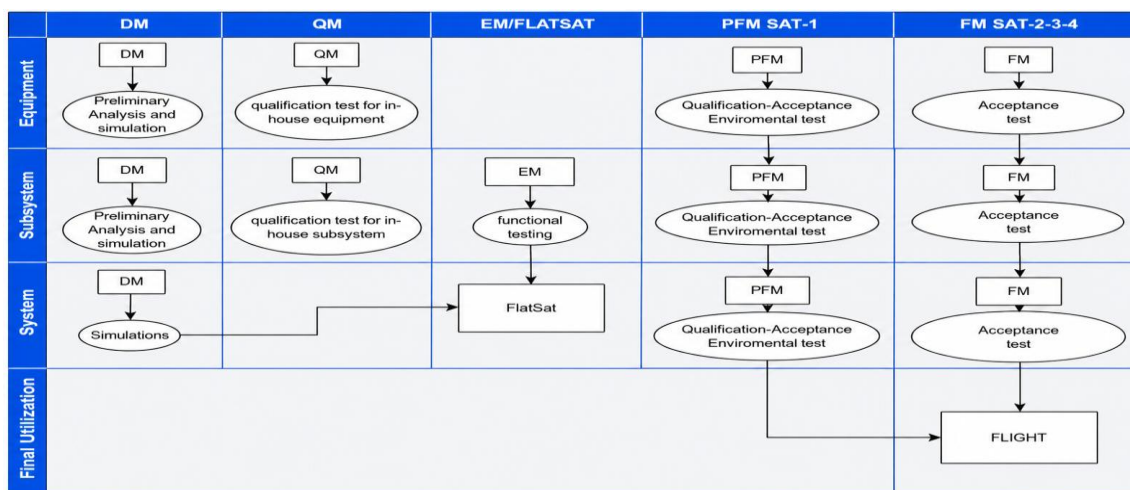


Figure 11-2: Model Philosophy

The trade-off results presented in Figure 11-2 indicate that the Hybrid Model Philosophy is the most suitable solution for the mission, as it provides an effective balance between cost, schedule, verification robustness, risk reduction, and implementation effort.

The selected model philosophy is implemented through a progressive sequence of models and verification activities:

- **Digital Models (DMs) and simulators:** These are used during the early design phases to support preliminary analyses, simulations, and the verification of functions that are difficult to reproduce under ground conditions.
- **Engineering Models (EMs):** These models are used to verify the main subsystems at hardware level, together with their electrical, functional, and communication interfaces, before final integration.
- **FlatSat:** The FlatSat serves as a test bench for the early verification of inter-subsystem communications, data handling, telemetry and telecommand functions, electrical interfaces, ADCS logic, EPS power distribution and payload operational sequences. This configuration allows critical issues to be identified and resolved before the final mechanical integration phase.
- **Qualification Models (QMs):** QMs, or equivalent qualification or delta-qualification activities, are foreseen for the internally developed payloads, as these are considered mission-critical elements.
- **Protoflight Model (PFM):** The first satellite in the constellation is used as the PFM and undergoes the complete functional and environmental test campaign.

The remaining three satellites are treated as recurring units and are subjected only to the acceptance tests required for Flight Models (FMs). This approach reduces both schedule and cost while maintaining the required level of reliability.

11.1.3 Verification Timeline

The verification, assembly, and integration plan, including the associated test activities, was developed in accordance with [AD06] and aligned with the mission timeline presented in Figure 11-3.

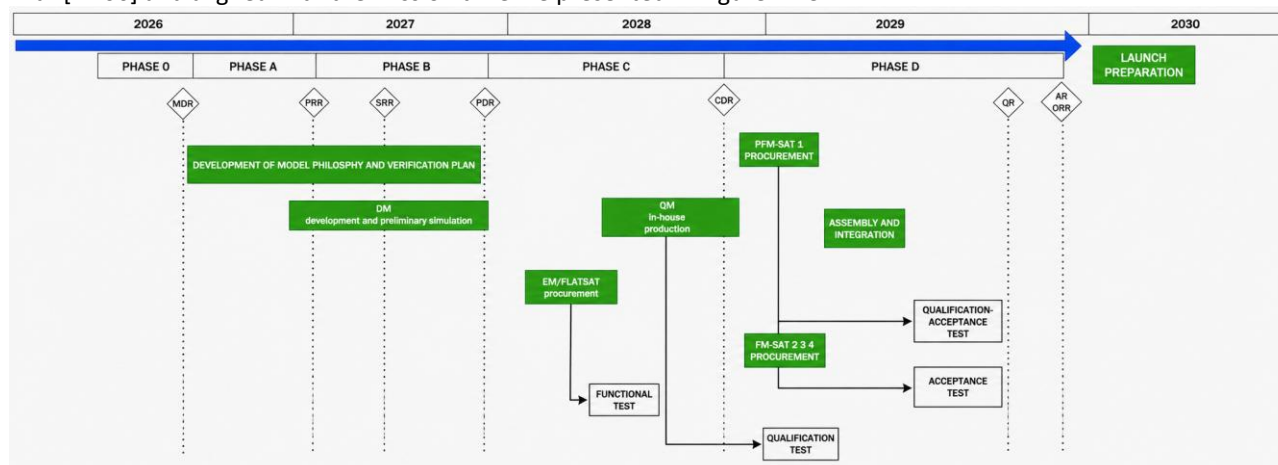


Figure 11-3: AIV Timeline

From the end of Phase 0 through the PDR, the primary effort is focused on defining the AIV strategy. In parallel, the first Digital Models (DMs) are developed and used for preliminary analyses and simulations supporting the initial design activities. Starting in Phase C, the preliminary development of the Qualification Models (QMs) intended for payload qualification is initiated, as the payloads are considered mission-critical elements. Development continues through manufacturing and the associated qualification testing in Phase D. During the same period, the procurement of the Engineering Models (EMs) and the execution of the corresponding preliminary functional tests are also planned. As the project progresses into Phase D, the procurement of the PFM and the subsequent FMs is planned, followed by assembly and integration activities and the execution of the tests required prior to launch preparation.

11.2 Capella Implementation

11.2.1 MBSE Workflow Integration, Implementation, and Developed Analyses

To enhance the efficiency and rigor of the system analysis and design processes required for the SmallSat Lunar Mission, Model-Based Systems Engineering (MBSE) approach was adopted using Capella. Through the Arcadia methodology, Capella allowed the team to replace a document-centric approach with an integrated, model-centric repository that strengthened consistency, traceability, and architectural coherence. This tool supported the engineering process by serving as a central, reliable reference model for all system definitions, allowing the concurrent design of the spacecraft while continuously verifying its logical and physical integrity. To streamline the engineering process and optimize time, the team adapted the standard workflow presented in Figure 11-4.

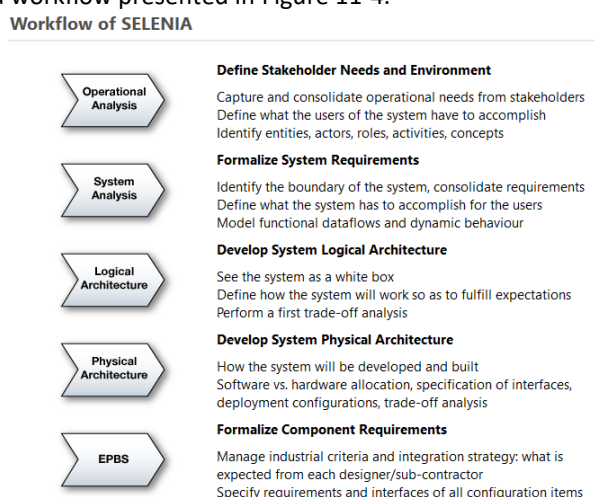


Figure 11-4: Workflow according to Capella

It is important to note that the team intentionally bypassed the explicit modelling of stakeholder needs within Capella, both to accelerate the initialization phase of the project and to ensure a more balanced and efficient distribution of the workload. The design process therefore began directly with the development of a comprehensive System Functional Breakdown (Functional Tree), produced through the Systems & Functional Analysis. This step enabled the team to decompose and structure the high-level mission objectives, such as characterizing the lunar radiation environment (LunaRAY) and monitoring biological responses (LunaCELL), into detailed system functions that directly informed the definition of subsystem requirements, interfaces, and operational behaviours. Concurrently, the team performed functional coverage and traceability analyses.

Transitioning to the Logical Architecture phase, and in accordance with the project workflow, the system was analysed as a conceptual “white box” to define how it would work to fulfil the requirements. This intermediate phase served to systematically bridge the functional breakdown with its physical implementation by validating the logical grouping of functions before committing to hardware constraints. Through the allocation of each function to a specific component, Capella’s automated validation rules enabled the team to perform architectural coherence checks. This ensured that all required mission functions were correctly mapped to physical components and that no architectural elements remained without a clearly justified operational role.

Moving to the Physical Architecture & EPBS phase defined how the system would be built, guiding the specification of physical and electrical interfaces. In this phase, the Subsystems Architecture diagram (Section 10.3, Figure 10-2) was produced, mapping the satellite's component network and defining its power and communication paths, including the regulated 28 V Power Bus, the Payload Data Bus, and the main Data Bus interconnecting all major subsystems. This layer supported interface-level verification by enabling checks of signal compatibility and power routing across the spacecraft, ensuring correct voltage allocation and preventing hardware mismatches. Finally, the integration strategy and component requirements directly yielded the spacecraft's Product Tree (Section 10.3, Figure 10-3), establishing the physical breakdown structure used for mass and power budgeting.

11.2.2 Strengths and Issues of Capella

The adopted MBSE through Capella offered some significant advantages that made the design process much more intuitive. One of the main strengths is the tool's ability to handle automated transitions between the different modelling phases, which made the workflow move much more smoothly. Instead of having to manually rebuild or transfer data at each step, it's possible to easily carry the previous work over from the initial functional analysis down to the physical implementations. Furthermore, because Capella is entirely based on visual schemes and graphical diagrams, it made it easy to visualize and understand the overall satellite architecture.

However, the engineering experience also highlighted a few noticeable downsides and practical challenges. Capella was a new software to the team and a complex system that requires a steep learning curve and a significant amount of time to master. Additionally, the tool's extensive modelling formalisms and strict workflow structure sometimes limited flexibility during rapid design iterations, making it harder to explore alternative concepts or adjust the architecture quickly when new ideas emerged.

12 Study management

12.1 Work Breakdown Structure

During the preliminary design phase, not all activities could be performed in parallel, as several disciplines depended on common mission-level inputs. Therefore, a first WBS was defined for Phase 0, where the team worked collaboratively to establish the mission foundations, including the Concept of Operations, system architecture, and orbit definition. For the reasons explained, in WPs of phase 0 all team members are considered participants.



Figure 12-1: WBS Phase 0

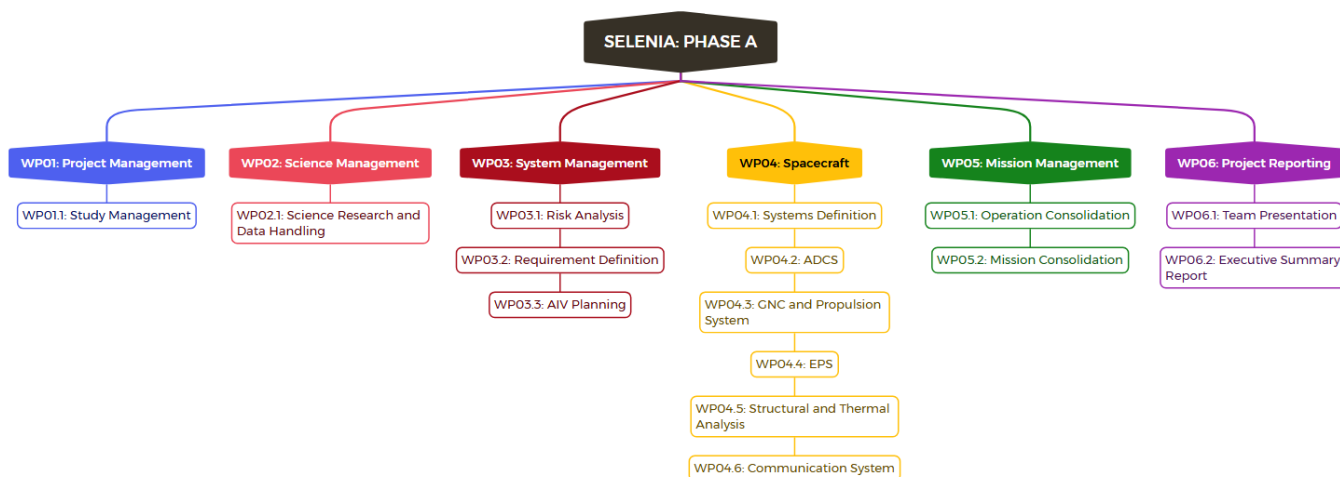


Figure 12-2: WBS Phase A

Following the architecture trade-off and selection process, the project entered Phase A. At this stage, activities could be distributed among the different disciplines, leading to the adoption of a second WBS focused on design consolidation and subsystem-level development.

A detailed description of all work packages is provided in Table 12-1, which reports, for each WP, the associated objectives and the corresponding list of tasks. The table covers the full scope of the study, from the initial Phase 0 activities including: teambuilding, mission objectives definition, and mission concept selection through the technical and management work packages of Phase A, namely Science Management, System Management, Spacecraft, Mission Management, Study Management, and Project Reporting.

Table 12-1: Work Package Description

WP	Type	Description
Teambuilding (0.00)	Objectives	Create team SELENIA and assign roles to all members.
	Task	Creation of project team Creation of communication and sharing channels Creation of OBS Creation of team's name and logo
Mission Objectives (0.01)	Objectives	Define stakeholders, their needs and objectives, values, QFD matrix and mission statement.
	Task	Stakeholder identification Power-interest matrix Stakeholder needs and objectives QFD matrix Mission Statement
Mission Definition (0.02)	Objectives	Define the mission concept through trade-off analysis and architecture discussions.
	Task	Mission concepts discussion Mission architectures discussion Trade-off study for defining our mission concept Project phase 0 reporting
Study Management (A.01)	Objectives	Coordinate all project activities in terms of schedule, deliverables, documentation, and meetings.
	Task	Coordination of the team Project control (schedule, deliverables and WPs) Progress reporting coordination Interface with Advisory Board Documentation and configuration control Preparation and/or review of final deliverables
Science Management (A.02)	Objectives	Select the biological species and define science requirements and data budget in compliance with mission needs.
	Task	Select biological species Identify payload operative modes Define data budget (with ComSys Engineer) Produce Science Requirements Build scientific foundation for System and Subsystem Engineers
System Management (A.03)	Objectives	Ensure technical consistency by coordinating system engineering, managing requirements, and defining the AIV approach.
	Task	Define project, mission, system and subsystem requirements Define the Science Traceability Matrix Define Model Philosophy Identify risks and implement mitigation plan
Spacecraft (A.04)	Objectives	Define spacecraft architecture, size all subsystems and produce the CAD model.
	Task	Define system and subsystem requirements Define functional tree and operative modes Size subsystems Define system architecture and interfaces Create spacecraft CAD model
Mission Management (A.05)	Objectives	Define orbit, ΔV budget, C3 architecture, and operative strategies; provide data to subsystem engineers.
	Task	Define final orbit in STK & ΔV budget Provide orbital data to subsystem engineers Define mission and operation requirements Define operative strategies Develop mission timeline Define C3 architecture and OBC computer architecture
Project Reporting (A.06)	Objectives	Production of all project deliverables, starting from individual reporting.
	Task	Create Team Presentation Select speakers Write Executive Summary Report

12.2 Study organisation and timeline

The KOM and SOW provided the initial mission drivers, which the Requirements Engineer used to compile a high-level requirements list. This activity, carried out jointly with Science Management, was initiated only after the mission statement had been established. With the mission statement defined, the Science Manager was able to identify the set of microorganisms best suited to the project objectives. Drawing on the five most significant values from the QFD matrix and on data regarding the preferred environmental conditions for each species, the team selected the mission concept and architecture, culminating in the Mission Definition Review (MDR), a formal milestone at which the final ConOps and mission architecture were agreed upon. The Study logic and work sequence diagram is reported in **Error! Reference source not found.** figure 12-3.

During the trade-off phase, Science Management and Systems Management worked in parallel, with the latter conducting a preliminary risk and requirements assessment concurrently with concept selection. At the close of Phase 0, all team members transitioned to their respective work packages: System Engineers first developed the functional tree, after which Spacecraft Engineers and Mission Management could begin their individual tasks.

The MDR was held on 21 April 2026 and formally marked the boundary between Phase 0 and Phase A. Note that in the Gantt chart (**Error! Reference source not found.** figure 12-4), Phase A is shown as beginning slightly before 22/04/2026; this is intentional, as it simplifies the Phase 0 WBS by keeping it at a higher level of abstraction.

The following key observations apply to the study logic and timeline:

1. Science Management was initiated during the mission concept trade-off to supply species requirements data and support a more informed selection process.
2. Risk analysis and requirements definition began following the collection of information from the SOW and the mission statement.
3. AIV planning was introduced only after a preliminary requirements baseline had been established.
4. Systems Definition started earlier than all other tasks, as the functional tree was a prerequisite for subsystem engineers to understand the scope and boundaries of their respective design activities.

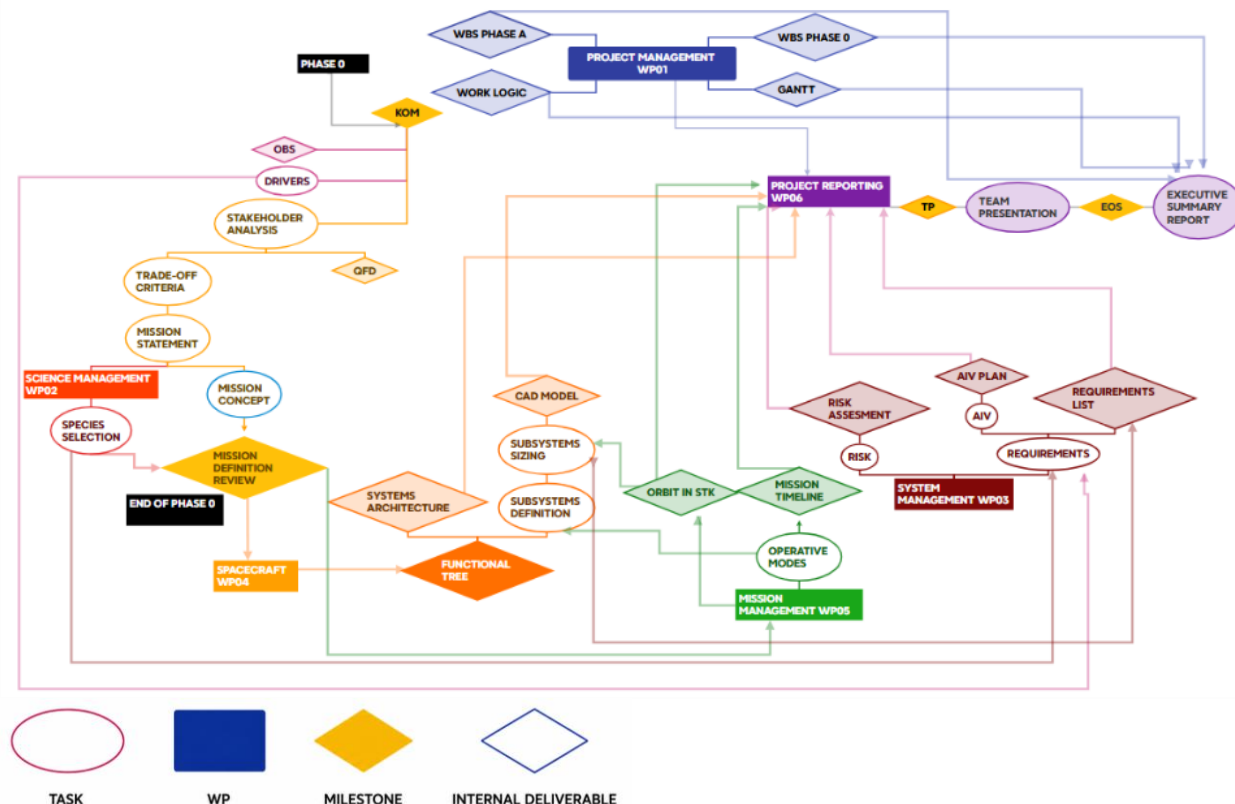


Figure 12-3: Work Logic

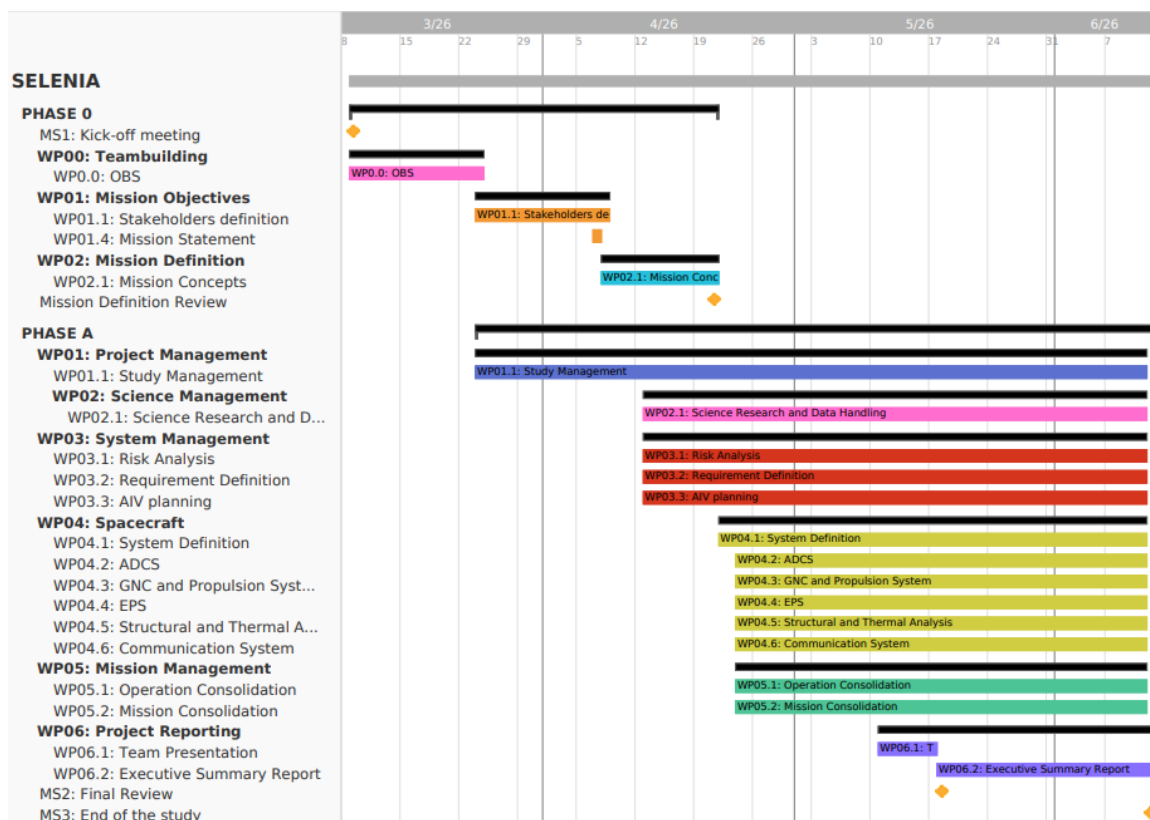


Figure 12-4: SELENIA GANTT Chart

12.3 Team organisation

The team organisational structure is illustrated in Figure 12-5. The team is subdivided into three main branches: Study Management, responsible for overall coordination and science activities; System Engineering, further divided into System Management, System Engineers, and the Spacecraft Unit covering all technical subsystems; and Mission Definition, responsible for trajectory design and operations. A total of 14 members is distributed across the three branches, each assigned to a specific role. A brief biography of each team member is reported in [Error! Reference source not found.-2](#).

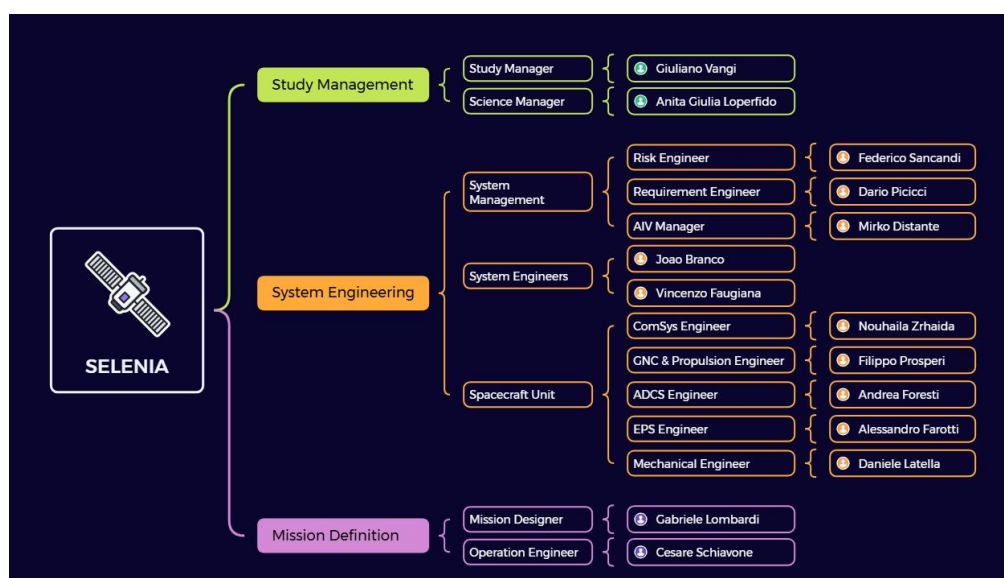
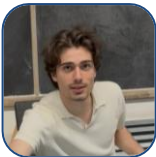


Figure 12-5: Organization Breakdown Structure

Table 12-2: Team Members Biography

 Giuliano Vangi: Study Manager MSc Aerospace Engineering student at Politecnico di Torino, interested in CFD and aerothermodynamics.	 Mirko Distante: AIV Manager MSc Aerospace Engineering student at Politecnico di Torino. Passionate about space exploration and Space Propulsion
 Anita Giulia Loperfido: Science Manager MSc Aerospace Engineering student at Politecnico di Torino and member of OSIRIS team as Propulsion engineer. Passionate about deep space exploration and scientific instrumentation, with interests in optical payloads and planetary science.	 Alessandro Farotti: EPS engineer MSc Aerospace Engineering at Politecnico di Torino. Passionate about space exploration and space propulsion, with interests in astronomy and astrophysics
 Cesare Schiavone: Operations Engineer MSc Aerospace Engineering student at Politecnico di Torino. Developed Mission Operations strategy, plan and timeline. Designed on-board Data Handling and Modes of Operations.	 Nouhaila Zrhaida: CommSys Engineer MSc Aerospace Engineering student at Politecnico di Torino and member of the CubeSat team as Thermal Engineer. Passionate about astronomy and orbital mechanics.
 Gabriele Lombardi: Mission Designer MSc Space Engineering student at Politecnico di Torino, with a strong interest in astrodynamics, mission analysis, and GNC systems. Particularly motivated by orbital mechanics and trajectory design for space missions.	 João Branco: System Engineer MSc student in aeronautical engineering at the University of Beira Interior, currently on Erasmus at Politecnico di Torino. Interested in satellite design and systems engineering.
 Daniele Latella: Mechanical Engineer MSc Aerospace Engineering student at Politecnico di Torino, passionate about space exploration. Former head of the Mechanics Area for Team ISAAC.	 Vincenzo Faugiana: System Engineer MSc Aerospace Engineering student, passionate about human space exploration. Member of Zefiro Team Polito where he's a System Engineer.
 Filippo Prosperi: GNC and Propulsion Engineer MSc Aerospace Engineering student at Politecnico di Torino, passionate about space and space propulsion. Member of the PoliTo Rocket Team, where he works as a Test Engineer in the Propulsion Department.	 Federico Sancandi: Risk Engineer MSc Aerospace Engineering student at Politecnico di Torino, interested in astronomy and astrophysics, who would love to become an astronaut.
 Andrea Foresti: ADCS Engineer MSc Aerospace Engineering student at Politecnico di Torino, where he pursues his passion for space and ADCS/GNC systems. He is the Astrodynamics Section Head of the OSIRIS team, where he has worked on debris risk assessment and spaceflight mechanics.	 Dario Picicci: Requirements Engineer MSc aerospace engineering student, with passion for astrophysics and space systems, in particular electric propulsors. Former PoliTOrbital Team member, he presented the team project at ESA's Student Aerospace Challenge 2025.

12.4 Reporting and communication

The study resulted in two main deliverables: the Final Review presentation, issued on 20/05/2026, and this Executive Summary Report.

Additionally, some intermediate deliverables were produced throughout the study, as reported in Table 12-3.

Table 12-3: List of Internal Deliverable

Internal deliverable	Description
<i>OBS</i>	Distribution of Roles
<i>WBS phase 0</i>	WPs list and tasks of Phase 0
<i>WBS phase A</i>	WPs list and tasks of Phase A
<i>Work Logic</i>	Flow followed during the project
<i>GANTT</i>	Project Timeline
<i>QFD Matrix</i>	Relationships Between Stakeholders and Their Prioritized Needs
<i>System Architecture</i>	Connection of all subsystems to the power bus and their respective data bus
<i>CAD Model</i>	Assembly of all subsystems in SolidWorks
<i>Orbit in STK</i>	Propagation of transfer orbit, operative orbit and disposal orbit in the STK environment
<i>Mission Timeline</i>	Mission Concept of Operations
<i>Risk Assessment</i>	List of Risk registers produced by Risk Engineer
<i>AIV Plan</i>	AIV strategy, including model philosophy and Verification Matrix
<i>Requirements List</i>	All requirements listed, with the help of all engineers, by the Requirements Engineer

Internal communications took place on informal means, while a bi-weekly team study session was held on Wednesdays, to complete weekly presentations, and on Fridays, to continue working, discuss milestones achieved and review the previously completed work. All documents produced throughout the study, both formal and informal, were uploaded and stored on Google Drive, accessible by all team members. Any communication with the SMSD teaching staff were instead conducted through Microsoft Teams.

12.5 Risk assessment

The risk assessment was conducted through the following steps:

- Risk identification through brainstorming and what-if analysis
- Identification of the causes and effects related to each risk
- Risk magnitude assessment by evaluating the severity and likelihood of each risk
- Identification of risk reduction measure

This process led to the identification of three different categories of mission risks:

- Cost risks (C): events that can cause the project to go over the mission budget
- Schedule risks (SCH): events that can lead to delays and the rescheduling of the mission
- Technical risks (TCH): events that can threaten the success of the mission and its scientific return

To determine the severity and likelihood of the risks before and after implementing the mitigation strategies, the following tables were used as a reference:

Score	Likelihood	Likelihood of occurrence	Score	Severity	Severity of consequence
E	Maximum	Almost certain to occur (more than 1 in 10)	5	Catastrophic	Total loss of mission
D	High	Will occur frequently (1 in 10)	4	Critical	Loss of major mission objectives or critical degradation of performance
C	Medium	Will occur sometimes (1 in 100)	3	Major	Significant impact on the mission
B	Low	Will seldom occur (1 in 1000)	2	Minor	Minor reduction in performance or mission return
A	Minimum	Will almost never occur (less than 1 in 10000)	1	Negligible	No noticeable impact on the mission

Figure 12-6: Severity-probability

The identification of risks and their root causes was primarily conducted through brainstorming, what-if analyses and fault trees, as a tool to identify the most critical components within each subsystem. For top-level scenarios, the starting point was the analysis of mission objectives and requirements, while for each subsystem, we coordinated with the responsible team members to fully understand the main criticalities and potential failure modes.

Risk mitigation was mainly achieved through a fault-tolerant design approach. Specifically, the identified mitigation actions involved the implementation of redundancies and the allocation of adequate margins. Once the risk responses were defined and their magnitude re-evaluated, a post-mitigation risk matrix was generated to assess the effectiveness of the implemented reduction measures.

12.5.1 Project level risks

The major project level risk identified was the potential missed delivery of the Phase A final review. Mitigation actions included establishing a strict baseline schedule, allocating time margins, and enforcing weekly progress tracking. During the study, this risk partially manifested when minor communication issues and subsystem misalignments temporarily threatened the project schedule. The situation was successfully handled by immediately executing the planned mitigation measures: the team utilized mandatory concurrent design sessions to quickly resolve the misalignments, effectively absorbing the delays and reducing the residual risk to a minimum likelihood.

The detailed Risk Register for this scenario is provided in Figure 12-7:

Risk Register						
Project:	Selenia		Source:	Federico Sancandi	Selenia	
WBS Ref:	WP01.1/WP06		Assigned to:	Federico Sancandi	Selenia	
Date:	24/04/2026		Controlled by:			
Issue:	1.0		Approved by:			
Risk Scenario and Decision						
Risk No.	SCH-01	Risk scenario title:	Missed delivery of the phase A final review package			
Cause:	- Lack of active participation of key team members during critical design phases - Inaccurate estimation of the time required for system-level design - Inadequate communication and information flow between team members - Inefficient teamwork - Failure to adequately define and baseline mission requirements and deliverables					
Consequence:	Failure to successfully close Phase A, preventing the transition to Phase B and resulting in project schedule delays					
Likelihood (L)	Severity (S)	Risk Index (before)	Risk domain	Risk Decision	Status	
Medium	Major	C3	Schedule	Reduce	Open with action	
Risk Mitigation						
Risk reduction measure:		- Develop a detailed work logic and establish a baseline schedule with sufficient time margins for critical activities - Enforce weekly progress tracking meetings - Enforce mandatory attendance for concurrent design sessions to ensure continuous subsystem alignment - Ensure that all project changes and design updates are promptly communicated to every team member				
Verification means:		Weekly analysis of the project Gantt chart, tracking task completion rates against the baseline schedule				
Expected risk reduction:		Likelihood (L)	Severity (S)	Risk Index (after)		
		Minimum	Major	A3		
Action(s):		- Define the project schedule and gantt chart before design phase kickoff - Assign clear ownership and hard deadlines for each specific document of the final review report				

Figure 12-7: Risk Register

12.6 Cost assessment

To estimate the total amount of work performed during the project, the Team Manager created an Excel spreadsheet in which each team member recorded the duration of individual work sessions. The duration of group activities and meetings was tracked directly by the Team Manager. The cumulative working hours obtained from these records were then used to estimate the overall personnel cost.

In addition to labour costs, the total mission cost also includes the annual license fees associated with the software tools used throughout the project:

1. MATLAB: 3000 € (standard version + eventual specific additional toolbox)
2. Capella: 6000€
3. STK: 55.000€
4. SolidWorks: 4700€

Table 12-4: Cost Assessment

<i>Member</i>	<i>Role</i>	<i>Total [h]</i>	<i>Salary per h [€]</i>	<i>Total cost [€]</i>
Anita Giulia Loperfido	Science Manager	175	17	2975
Vincenzo Faugiana	System Engineer	167	12	2004
Joao Branco	System Engineer	150	12	1800
Giuliano Vangi	Study Manager	175	17	2975
Federico Sancandi	Risk Engineer	150	12	1800
Dario Picicci	Requirements engineer	164	12	1968
Mirko Distante	AIV Manager	150	17	2550
Nouhaila Zrhaida	COMSYS engineer	164	12	1968
Filippo Prosperi	GNC engineer	164	12	1968
Andrea Foresti	ADCS Engineer	167	12	2004
Alessandro Farotti	EPS engineer	162	12	2004
Daniele Latella	Mechanical Engineer	162	12	1944
Cesare Schiavone	Operations Engineer	155	12	1860
Gabriele Lombardi	Mission Engineer	155	12	1860
TOTAL				29.632€
TOTAL WITH LICENCES:				98.332€

To estimate costs, we considered the average salary for engineering and management entry-level positions in Italy. Some key aspects can be highlighted:

1. The first work package addressed by the team was WP01 (Phase 0). Its completion required a considerable amount of time, as the team members had not previously worked together and initially needed time to become familiar with each other's working styles and establish an effective collaboration. As a result, WP02 was completed more efficiently.
2. The workload was not evenly distributed throughout the project. The beginning of Phase A required the greatest effort, as a large amount of information was needed to define the mission architecture and perform the preliminary sizing of the subsystems. To support this process, each team member carried out research activities within their area of responsibility, providing the data necessary for the subsequent design phases.

13 Conclusions and recommendations

The Phase 0/A study demonstrated the technical feasibility of SELENIA and showed that the selected mission architecture can fulfil the scientific objectives while complying with the main mission requirements. During the design process, Team SELENIA deliberately chose to extend the mission duration beyond the minimum value imposed by the project drivers. Although this decision resulted in larger subsystem sizing and increased resource requirements, it was considered a worthwhile trade-off to maximize scientific return and provide the community with a more extensive dataset in both temporal and spatial coverage. The resulting architecture satisfies all mission drivers and requirements while ensuring increased operational flexibility. Due to conservative assumptions adopted during the design process, the mission lifetime ultimately exceeds the nominal 720-day science phase.

Several strengths emerged from the study. First, the selected microorganisms provide a strong scientific foundation, paired with dataset correlations between the two payloads. The proposed disposal strategy ensures compliance with planetary protection principles and supports the long-term sustainability goal of lunar operations. Moreover, the selected orbital configuration enables high-resolution characterization of the radiation environment and improved mapping of albedo particles over the lunar south polar regions. Finally, the effectiveness of the design process strongly benefited from the cooperative approach and interdisciplinary expertise of the team members. An approach that proved particularly effective was the active collaboration between team members beyond the responsibilities associated with their assigned roles. Supporting one another in specific tasks improved the overall consistency of the documentation and ensured continuity of the work in the event of a team member's absence.

Despite these positive results, some critical aspects were identified. The transfer duration remains longer than the recommended time to guarantee that the biological payload remains fully viable before the start of science operations. In addition, reliance on Commercial-Off-The-Shelf components limits radiation tolerance, highlighting the need for more dedicated hardware solutions. Further improvements could also be achieved by increasing subsystem-level redundancy and optimizing the thermal control strategy, particularly regarding power consumption.

Future developments should therefore focus on reducing the transfer time below 60 days to enable earlier payload activation, increasing the use of radiation-resistant technologies, and refining subsystem designs to improve reliability and robustness. Overall, SELENIA represents a valuable and scalable European contribution to lunar exploration and provides a solid basis for future studies aimed at supporting long-term human presence beyond Earth.